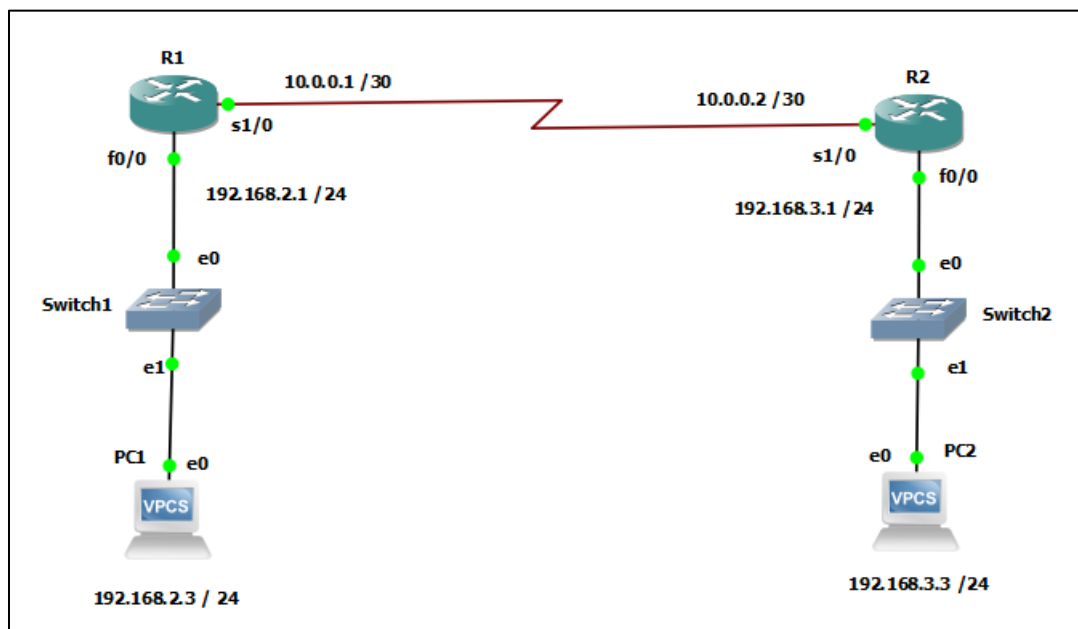


PRACTICAL 1A

AIM: Implement the concept of Static Routing:

TOPOLOGY:



STEPS:

Step 1: Configure Both Interfaces on R1 (f0/0 and s1/0)

```
R1#conf t
R1(config) #int fa0/0
R1(config-if) #ip address 192.168.2.1 255.255.255.0
R1(config-if) #no shut
R1(config-if) #exit
R1(config)#int s1/0
R1(config-if) #ip address 10.0.0.1 255.255.255.252
R1(config-if) #no shut
R1(config-if) #clock rate 64000
R1(config-if)#exit
```

```

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
*Mar  1 00:08:04.287: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:08:05.287: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R1(config-if)#exit

```

```

R1(config)#int s1/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#
*Mar  1 00:08:50.191: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R1(config-if)#
*Mar  1 00:08:51.195: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0,
changed state to up
R1(config-if)#clock rate 64000
R1(config-if)#exit

```

STEP 2: Configure both interfaces on R2 [f0/0 and s1/0]

```

R2#conf t
R2(config) #int fa0/0
R2(config-if) #ip address 192.168.3.1 255.255.255.0
R2(config-if) #no shut
R2(config-if) #exit
R2(config) #int s1/0
R2(config-if) #ip address 10.0.0.2 255.255.255.252
R2(config-if) #no shut
R2(config-if) #exit

```

for f0/0]

```

R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 192.168.3.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#
*Mar  1 00:08:17.283: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:08:18.283: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R2(config-if)#exit

```

For s1/0

```
R2(config)#int s1/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#
*Mar 1 00:08:57.683: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R2(config-if)#
*Mar 1 00:08:58.687: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0,
changed state to up
R2(config-if)#clock rate 64000
R2(config-if)#exit
```

STEP 3 : Use do show ip route command : shows all the networks the router knows and how to reach them.

R1(Config) # do show ip route

```
R1(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial1/0
C     192.168.2.0/24 is directly connected, FastEthernet0/0
```

R2 (Config) # do show ip route

```
R2(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial1/0
C     192.168.3.0/24 is directly connected, FastEthernet0/0
```

STEP 4: Configure Static Route on R1 and R2

R1(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.2

R1(config)#do show ip route

```
R1(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.2
R1(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial1/0
C       192.168.2.0/24 is directly connected, FastEthernet0/0
S       192.168.3.0/24 [1/0] via 10.0.0.2
```

R2(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.1

R2(config)#do show ip route

```
R2(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.1
R2(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial1/0
S       192.168.2.0/24 [1/0] via 10.0.0.1
C       192.168.3.0/24 is directly connected, FastEthernet0/0
```

STEP 5 : Manually Set an IP on PC1

PC1> ip 192.168.2.3/24 192.168.2.1

```
PC1> ip 192.168.2.3/24 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.3 255.255.255.0 gateway 192.168.2.1
```

STEP 6 : Manually Set an IP on PC2

PC2> ip 192.168.3.3/24 192.168.3.1

```
PC2> ip 192.168.3.3/24 192.168.3.1
Checking for duplicate address...
PC1 : 192.168.3.3 255.255.255.0 gateway 192.168.3.1
```

STEP 7: Ping each far network / Test Connectivity using ping

PC1> ping 192.168.3.3

```
PC1> ping 192.168.3.3
84 bytes from 192.168.3.3 icmp_seq=1 ttl=62 time=61.121 ms
84 bytes from 192.168.3.3 icmp_seq=2 ttl=62 time=60.647 ms
84 bytes from 192.168.3.3 icmp_seq=3 ttl=62 time=60.767 ms
84 bytes from 192.168.3.3 icmp_seq=4 ttl=62 time=61.067 ms
84 bytes from 192.168.3.3 icmp_seq=5 ttl=62 time=61.363 ms
```

PC2> ping 192.168.2.3

```
PC2> ping 192.168.2.3
84 bytes from 192.168.2.3 icmp_seq=1 ttl=62 time=60.509 ms
84 bytes from 192.168.2.3 icmp_seq=2 ttl=62 time=61.891 ms
84 bytes from 192.168.2.3 icmp_seq=3 ttl=62 time=60.555 ms
84 bytes from 192.168.2.3 icmp_seq=4 ttl=62 time=60.782 ms
84 bytes from 192.168.2.3 icmp_seq=5 ttl=62 time=60.950 ms
```

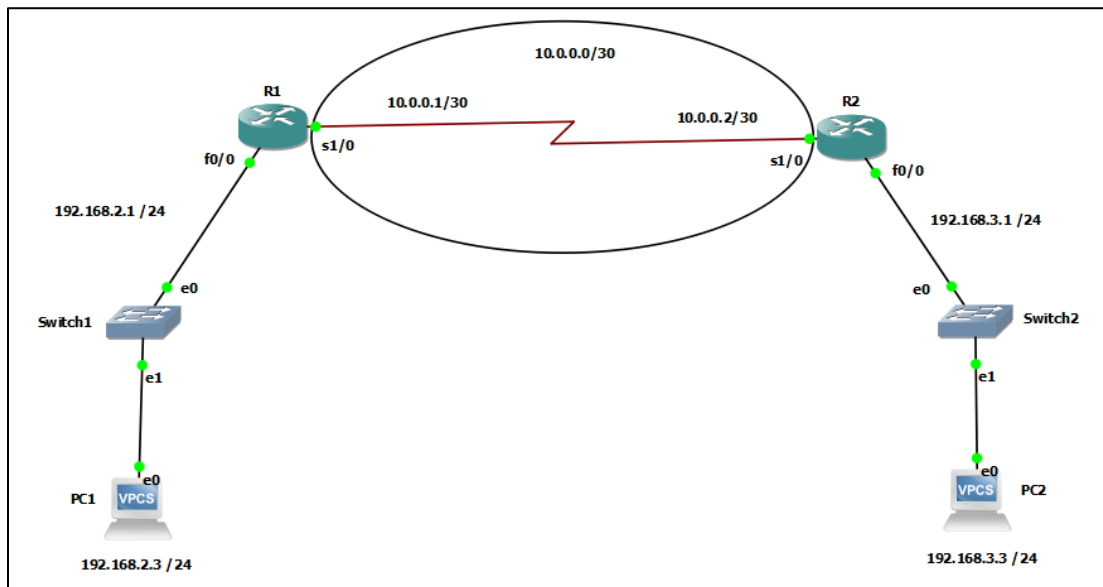
CONCLUSION:

The concept of Static Routing was implemented correctly as both PCs can ping each other successfully.

PRACTICAL 1B

AIM: Implement Dynamic Routing Protocol (RIP)

TOPOLOGY:



STEPS:

STEP 1: Configuring interfaces on R1

```
R1#conf t
R1(config)#int fa0/0
R1(config-if) #ip address 192.168.2.1 255.255.255.0
R1(config-if) #no shut
R1(config-if) #exit
R1(config) #int s1/0
R1(config-if) #ip address 10.0.0.1 255.255.255.252
R1(config-if) #no shut
R1(config-if) #exit
```

For f0/0

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
*Mar 1 00:00:41.891: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:00:42.891: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

for s1/0

```
R1(config)#int s1/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
*Mar 1 00:02:00.191: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R1(config)#
*Mar 1 00:02:01.195: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
```

STEP 2: Configure interfaces on R2

```
R2#conf t
R2(config)#int fa0/0
R2(config-if) #ip address 192.168.3.1 255.255.255.0
R2(config-if) #no shut
R2(config-if) #exit
R2(config) #int s1/0
R2(config-if) #ip address 10.0.0.2 255.255.255.252
R2(config-if) #no shut
R2(config-if) #exit
```

For f0/0

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 192.168.3.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar 1 00:03:21.803: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:22.803: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

For s1/0

```
R2(config)#int s1/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar  1 00:03:47.511: %LINK-3-UPDOWN: Interface Serial1/0, changed sta
R2(config)#
*Mar  1 00:03:48.515: %LINEPROTO-5-UPDOWN: Line protocol on Interface
changed state to up
```

STEP 3: Manually Set an IP on PC1 and PC2

PC1> ip 192.168.2.3/24 192.168.2.1

PC2> ip 192.168.3.3/24 192.168.3.1

```
PC1> ip 192.168.2.3/24 192.168.2.1
Checking for duplicate address...
PC1 : 192.168.2.3 255.255.255.0 gateway 192.1
```

```
PC2> ip 192.168.3.3/24 192.168.3.1
Checking for duplicate address...
PC1 : 192.168.3.3 255.255.255.0 gateway 192.168.3.1
```

STEP 4: Configuring RIP on R1 and show ip route

```
R1(config)#router rip
R1(config-router)#network 10.0.0.0
R1(config-router)#network 192.168.2.0
R1(config-router)#exit
R1(config)#do show ip route
```

```
R1(config)#router rip
R1(config-router)#network 10.0.0.0
R1(config-router)#network 192.168.2.0
R1(config-router)#exit
R1(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial1/0
C       192.168.2.0/24 is directly connected, FastEthernet0/0
```

STEP 5: Configuring RIP on R2 and show ip route

```
R2(config)#router rip
R2(config-router)#network 10.0.0.0
R2(config-router)#network 192.168.3.0
R2(config-router)#exit
```


R1(config) #do show ip route

```
R2(config)#router rip
R2(config-router)#network 10.0.0.0
R2(config-router)#network 192.168.3.0
R2(config-router)#exit
R2(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external t
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - I
        ia - IS-IS inter area, * - candidate default, U - per-use
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial1/0
R       192.168.2.0/24 [120/1] via 10.0.0.1, 00:00:13, Serial1/0
C       192.168.3.0/24 is directly connected, FastEthernet0/0
```

STEP 6: Ping the end PC's

PC1> ping 192.168.3.3

```
PC1> ping 192.168.3.3
84 bytes from 192.168.3.3 icmp_seq=1 ttl=62 time=60.516 ms
84 bytes from 192.168.3.3 icmp_seq=2 ttl=62 time=60.649 ms
84 bytes from 192.168.3.3 icmp_seq=3 ttl=62 time=61.021 ms
84 bytes from 192.168.3.3 icmp_seq=4 ttl=62 time=60.787 ms
84 bytes from 192.168.3.3 icmp_seq=5 ttl=62 time=61.639 ms
```

PC2> ping 192.168.2.3

```
PC2> ping 192.168.2.3
84 bytes from 192.168.2.3 icmp_seq=1 ttl=62 time=60.864 ms
84 bytes from 192.168.2.3 icmp_seq=2 ttl=62 time=61.567 ms
84 bytes from 192.168.2.3 icmp_seq=3 ttl=62 time=60.872 ms
84 bytes from 192.168.2.3 icmp_seq=4 ttl=62 time=60.808 ms
84 bytes from 192.168.2.3 icmp_seq=5 ttl=62 time=60.869 ms
```

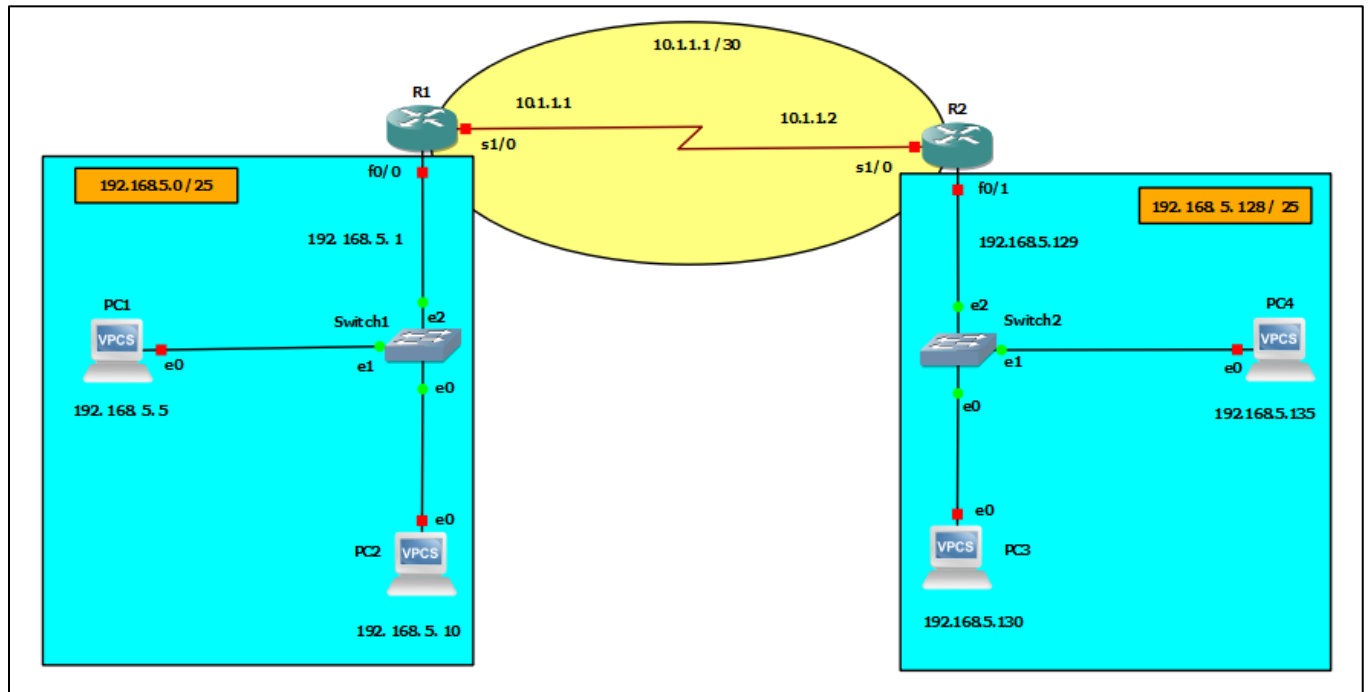
CONCLUSION:

RIP was configured correctly on both routers. Once the RIP networks were advertised, each router learned the other's LAN network dynamically. As a result, both PCs could successfully communicate, verifying the correct operation of the RIP routing protocol.

PRACTICAL 2A

AIM: Implement the concept of RIPv1 and RIPv2 routing protocol

TOPOLOGY:



STEPS :

STEP 1: Configure VPCSs

PC1> ip 192.168.5.5 /25 192.168.5.1

PC1> show ip

```
PC1> ip 192.168.5.5/25 192.168.5.1
Checking for duplicate address...
PC1 : 192.168.5.5 255.255.255.128 gateway 192.168.5.1

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.5.5/25
GATEWAY    : 192.168.5.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10022
RHOST:PORT : 127.0.0.1:10023
MTU        : 1500
```

PC2> ip 192.168.5.10 /25 192.168.5.1

PC2> show ip

```
PC2> ip 192.168.5.10/25 192.168.5.1
Checking for duplicate address...
PC1 : 192.168.5.10 255.255.255.128 gateway 192.168.5.1

PC2> show ip

NAME           : PC2[1]
IP/MASK        : 192.168.5.10/25
GATEWAY        : 192.168.5.1
DNS            :
MAC            : 00:50:79:66:68:01
LPORT          : 10024
RHOST:PORT     : 127.0.0.1:10025
MTU:           : 1500
```

PC3> ip 192.168.5.130 /25 192.168.5.129

PC3> show ip

```
PC3> ip 192.168.5.130/25 192.168.5.129
Checking for duplicate address...
PC1 : 192.168.5.130 255.255.255.128 gateway 192.168.5.129

PC3> show ip

NAME           : PC3[1]
IP/MASK        : 192.168.5.130/25
GATEWAY        : 192.168.5.129
DNS            :
MAC            : 00:50:79:66:68:02
LPORT          : 10026
RHOST:PORT     : 127.0.0.1:10027
MTU:           : 1500
```

PC4> ip 192.168.5.135 /25 192.168.5.129

PC4> show ip

```
PC4> ip 192.168.5.135 /25 192.168.5.129
Checking for duplicate address...
PC1 : 192.168.5.135 255.255.255.128 gateway 192.168.5.129

PC4> show ip

NAME           : PC4[1]
IP/MASK        : 192.168.5.135/25
GATEWAY        : 192.168.5.129
DNS            :
MAC            : 00:50:79:66:68:03
LPORT          : 10028
RHOST:PORT     : 127.0.0.1:10029
MTU:           : 1500
```

STEP 2: Configure the Ports on R1

for f0/0

```
R1(config)#int f0/0
R1(config-if)#ip address 192.168.5.1 255.255.255.128
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
*Mar  1 00:12:20.919: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:12:21.919: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

For s1/0

```
R1(config)#int s1/0
R1(config-if)#ip address 10.1.1.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#end
R1#
*Mar  1 00:13:42.039: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Type command: **do show ip / show interface brief**

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status  Prot
ocol
FastEthernet0/0        192.168.5.1     YES manual up      up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial1/0               10.1.1.1        YES manual up      down
Serial1/1               unassigned      YES unset  administratively down down
Serial1/2               unassigned      YES unset  administratively down down
Serial1/3               unassigned      YES unset  administratively down down
```

STEP 3: Configure the Ports on R2

For f0/0

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/1
R2(config-if)#ip address 192.168.5.129 255.255.255.128
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar  1 00:19:53.539: %LINK-3-UPDOWN: Interface FastEthernet0/1,
o up
```

For s1/0

```
R2(config)#int s1/0
R2(config-if)#ip address 10.1.1.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Mar  1 00:20:29.515: %LINK-3-UPDOWN: Interface Serial1/0, c
R2(config)#
*Mar  1 00:20:30.519: %LINEPROTO-5-UPDOWN: Line protocol on
```

Type command: show ip interface brief

```
R2(config)#do show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	192.168.5.129	YES	manual	up	up
Serial1/0	10.1.1.2	YES	manual	up	up
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down

STEP 4: Ping the serial interface of R2 from PC1:

PC1> ping 10.1.1.2 Connection will fail cause there is no suitable routing path between two devices rest other will ping successfully.

```
PC1> ping 192.168.5.10
84 bytes from 192.168.5.10 icmp_seq=1 ttl=64 time=2.152 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=64 time=1.437 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=64 time=1.546 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=64 time=1.496 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=64 time=1.737 ms

PC1> ping 192.168.5.1
84 bytes from 192.168.5.1 icmp_seq=1 ttl=255 time=15.274 ms
84 bytes from 192.168.5.1 icmp_seq=2 ttl=255 time=14.796 ms
84 bytes from 192.168.5.1 icmp_seq=3 ttl=255 time=16.197 ms
84 bytes from 192.168.5.1 icmp_seq=4 ttl=255 time=15.437 ms
84 bytes from 192.168.5.1 icmp_seq=5 ttl=255 time=15.556 ms

PC1> ping 10.1.1.1
84 bytes from 10.1.1.1 icmp_seq=1 ttl=255 time=15.278 ms
84 bytes from 10.1.1.1 icmp_seq=2 ttl=255 time=15.614 ms
84 bytes from 10.1.1.1 icmp_seq=3 ttl=255 time=15.886 ms
84 bytes from 10.1.1.1 icmp_seq=4 ttl=255 time=15.591 ms
84 bytes from 10.1.1.1 icmp_seq=5 ttl=255 time=15.345 ms

PC1> ping 10.1.1.2
10.1.1.2 icmp_seq=1 timeout
10.1.1.2 icmp_seq=2 timeout
10.1.1.2 icmp_seq=3 timeout
10.1.1.2 icmp_seq=4 timeout
10.1.1.2 icmp_seq=5 timeout
```

STEP 4: Ping the serial interface of R1 from PC3:

PC3> **ping 10.1.1.1** Connection will fail cause there is no suitable routing path between two devices rest other will ping successfully

```
PC3> ping 192.168.5.135
84 bytes from 192.168.5.135 icmp_seq=1 ttl=64 time=1.637 ms
84 bytes from 192.168.5.135 icmp_seq=2 ttl=64 time=1.580 ms
84 bytes from 192.168.5.135 icmp_seq=3 ttl=64 time=1.531 ms
84 bytes from 192.168.5.135 icmp_seq=4 ttl=64 time=1.467 ms
84 bytes from 192.168.5.135 icmp_seq=5 ttl=64 time=1.510 ms

PC3> ping 192.168.5.129
84 bytes from 192.168.5.129 icmp_seq=1 ttl=255 time=15.950 ms
84 bytes from 192.168.5.129 icmp_seq=2 ttl=255 time=15.514 ms
84 bytes from 192.168.5.129 icmp_seq=3 ttl=255 time=15.447 ms
84 bytes from 192.168.5.129 icmp_seq=4 ttl=255 time=15.559 ms
84 bytes from 192.168.5.129 icmp_seq=5 ttl=255 time=15.983 ms

PC3> ping 10.1.1.2
84 bytes from 10.1.1.2 icmp_seq=1 ttl=255 time=15.832 ms
84 bytes from 10.1.1.2 icmp_seq=2 ttl=255 time=15.503 ms
84 bytes from 10.1.1.2 icmp_seq=3 ttl=255 time=15.274 ms
84 bytes from 10.1.1.2 icmp_seq=4 ttl=255 time=16.067 ms
84 bytes from 10.1.1.2 icmp_seq=5 ttl=255 time=15.480 ms

PC3> ping 10.1.1.1
10.1.1.1 icmp_seq=1 timeout
10.1.1.1 icmp_seq=2 timeout
10.1.1.1 icmp_seq=3 timeout
10.1.1.1 icmp_seq=4 timeout
10.1.1.1 icmp_seq=5 timeout
```

STEP 5: Enable RIPv2 on Router 1

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 10.1.1.0
R1(config-router)#network 192.168.5.0
R1(config-router)#end
R1#
```

STEP 6: Enable RIPv2 on Router 2

```
R2#conf t
Enter configuration commands, one per line.  End with C
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#network 10.1.1.0
R2(config-router)#network 192.168.5.128
R2(config-router)#end
R2#
*Mar  1 00:43:24.767: %SYS-5-CONFIG_I: Configured from
```

STEP 7: Again Ping the serial interface of R2 from PC1:

PC1 will successfully ping the R2 Serial interface

```

PC1> ping 10.1.1.2
84 bytes from 10.1.1.2 icmp_seq=1 ttl=254 time=46.064 ms
84 bytes from 10.1.1.2 icmp_seq=2 ttl=254 time=45.746 ms
84 bytes from 10.1.1.2 icmp_seq=3 ttl=254 time=45.879 ms
84 bytes from 10.1.1.2 icmp_seq=4 ttl=254 time=45.997 ms
84 bytes from 10.1.1.2 icmp_seq=5 ttl=254 time=45.587 ms

PC1> ping 192.168.5.130
84 bytes from 192.168.5.130 icmp_seq=1 ttl=62 time=60.597 ms
84 bytes from 192.168.5.130 icmp_seq=2 ttl=62 time=60.696 ms
84 bytes from 192.168.5.130 icmp_seq=3 ttl=62 time=61.473 ms
84 bytes from 192.168.5.130 icmp_seq=4 ttl=62 time=61.012 ms
84 bytes from 192.168.5.130 icmp_seq=5 ttl=62 time=61.152 ms

PC1> ping 192.168.5.135
84 bytes from 192.168.5.135 icmp_seq=1 ttl=62 time=61.604 ms
84 bytes from 192.168.5.135 icmp_seq=2 ttl=62 time=60.649 ms
84 bytes from 192.168.5.135 icmp_seq=3 ttl=62 time=60.442 ms
84 bytes from 192.168.5.135 icmp_seq=4 ttl=62 time=59.729 ms
84 bytes from 192.168.5.135 icmp_seq=5 ttl=62 time=60.412 ms

```

STEP 8 : Again Ping the serial interface of R1 from PC3:

PC3 will successfully ping the R1 Serial interface

```

PC3> ping 10.1.1.1
84 bytes from 10.1.1.1 icmp_seq=1 ttl=254 time=46.711 ms
84 bytes from 10.1.1.1 icmp_seq=2 ttl=254 time=45.165 ms
84 bytes from 10.1.1.1 icmp_seq=3 ttl=254 time=45.898 ms
84 bytes from 10.1.1.1 icmp_seq=4 ttl=254 time=46.292 ms
84 bytes from 10.1.1.1 icmp_seq=5 ttl=254 time=46.137 ms

PC3> ping 192.168.5.5
84 bytes from 192.168.5.5 icmp_seq=1 ttl=62 time=60.518 ms
84 bytes from 192.168.5.5 icmp_seq=2 ttl=62 time=61.324 ms
84 bytes from 192.168.5.5 icmp_seq=3 ttl=62 time=60.658 ms
84 bytes from 192.168.5.5 icmp_seq=4 ttl=62 time=61.410 ms
84 bytes from 192.168.5.5 icmp_seq=5 ttl=62 time=61.043 ms

PC3> ping 192.168.5.10
84 bytes from 192.168.5.10 icmp_seq=1 ttl=62 time=61.253 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=62 time=61.361 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=62 time=60.784 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=62 time=61.998 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=62 time=61.009 ms

```

STEP 9: Verify RIPv2

R1# show ip rip database

R1# show ip route

```

R1#show ip rip database
10.0.0.0/8      auto-summary
10.1.1.0/30     directly connected, Serial1/0
192.168.5.0/24  auto-summary
192.168.5.0/24
    [1] via 10.1.1.2, 00:00:21, Serial1/0
192.168.5.0/25  directly connected, FastEthernet0/0

```



```

R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.5.0/25 is directly connected, FastEthernet0/0
R       192.168.5.0/24 [120/1] via 10.1.1.2, 00:00:25, Serial1/0
      10.0.0.0/30 is subnetted, 1 subnets
C       10.1.1.0 is directly connected, Serial1/0

```

Do the same for R2 as well:

R2# show ip rip database

R2# show ip route

```

R2#show ip rip database
10.0.0.0/8      auto-summary
10.1.1.0/30     directly connected, Serial1/0
192.168.5.0/24  auto-summary
192.168.5.0/24
    [1] via 10.1.1.1, 00:00:30, Serial1/0
192.168.5.128/25 directly connected, FastEthernet0/1

```

```

R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
R       192.168.5.0/24 [120/1] via 10.1.1.1, 00:00:08, Serial1/0
C       192.168.5.128/25 is directly connected, FastEthernet0/1
      10.0.0.0/30 is subnetted, 1 subnets
C       10.1.1.0 is directly connected, Serial1/0

```

STEP 10: Review RIPv2 Messages

Use “**debug ip rip**” command for both routers to check RIPv2 configuration

```

R1#debug ip rip
RIP protocol debugging is on
R1#
*Mar  1 01:36:47.295: RIP: received v2 update from 10.1.1.2 on Serial1/0
*Mar  1 01:36:47.299:      192.168.5.0/24 via 0.0.0.0 in 1 hops
R1#
*Mar  1 01:36:54.967: RIP: sending v2 update to 224.0.0.9 via FastEthernet0/0 (192.168.5.1)
*Mar  1 01:36:54.967: RIP: build update entries
*Mar  1 01:36:54.967:   10.0.0.0/8 via 0.0.0.0, metric 1, tag 0
*Mar  1 01:36:54.971:   192.168.5.0/24 via 0.0.0.0, metric 2, tag 0
R1#
*Mar  1 01:36:57.003: RIP: sending v2 update to 224.0.0.9 via Serial1/0 (10.1.1.1)
*Mar  1 01:36:57.003: RIP: build update entries
*Mar  1 01:36:57.003:   192.168.5.0/24 via 0.0.0.0, metric 1, tag 0

```



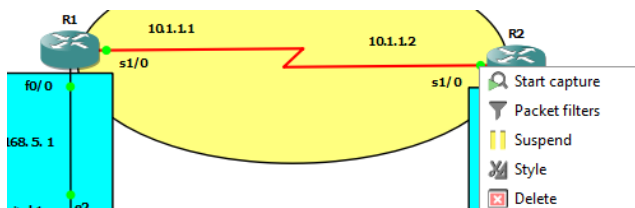
```

R2#debug ip rip
RIP protocol debugging is on
R2#
*Mar  1 01:29:16.703: RIP: sending v2 update to 224.0.0.9 via Serial1/0 (10.1.1.2)
*Mar  1 01:29:16.703: RIP: build update entries
*Mar  1 01:29:16.703:   192.168.5.0/24 via 0.0.0.0, metric 1, tag 0
R2#
*Mar  1 01:29:23.095: RIP: received v2 update from 10.1.1.1 on Serial1/0
*Mar  1 01:29:23.095:   192.168.5.0/24 via 0.0.0.0 in 1 hops
R2#
*Mar  1 01:29:24.995: RIP: sending v2 update to 224.0.0.9 via FastEthernet0/1 (192.168.5.129)
*Mar  1 01:29:24.995: RIP: build update entries
*Mar  1 01:29:24.995:   10.0.0.0/8 via 0.0.0.0, metric 1, tag 0
*Mar  1 01:29:24.999:   192.168.5.0/24 via 0.0.0.0, metric 2, tag 0

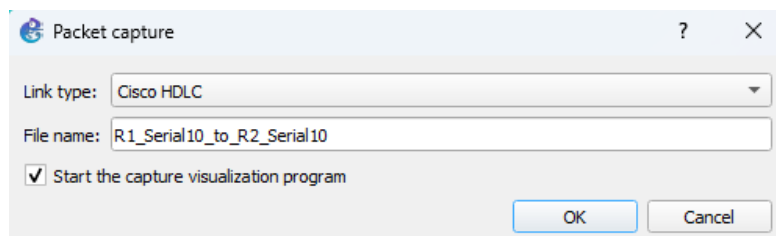
```

STEP 11 : Using Wireshark to Verify RIPv2

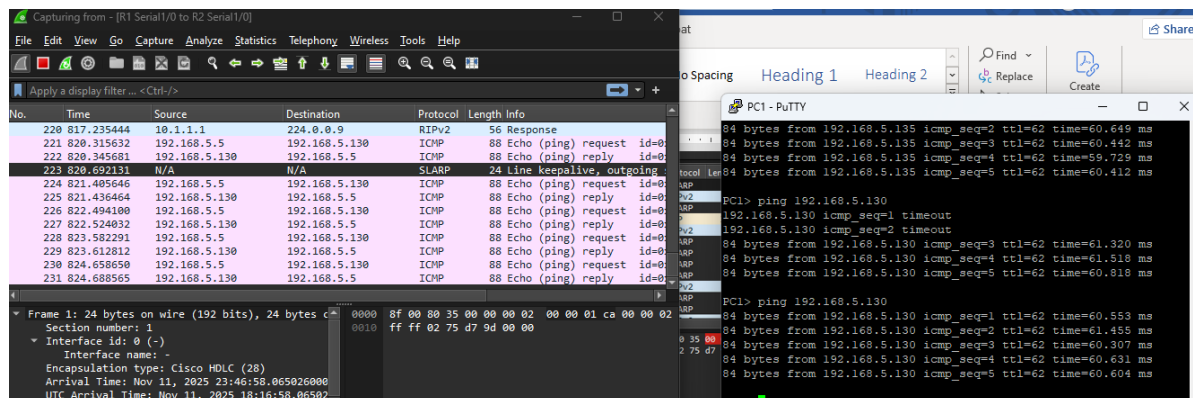
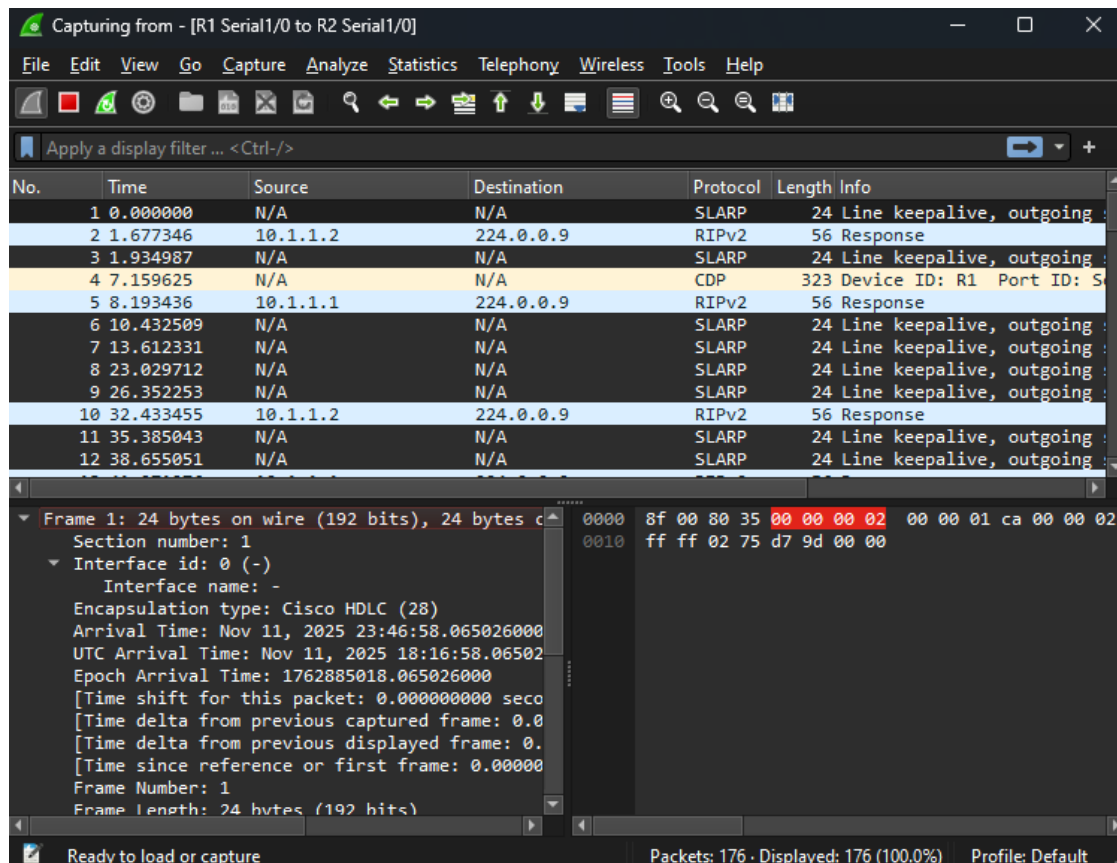
Right-click Serial link between R1 → R2 → Select “Start Capture” → Observe routing protocol activity



Open Packet Capture → Verify Link Type = Cisco HDLC → Click OK → View RIP Version 2 details



When Wireshark Starts → Observe RIP Version 2 packets → Note multicast address 224.0.0.9 → Understand RIPv2 uses this address to send routing updates between routers



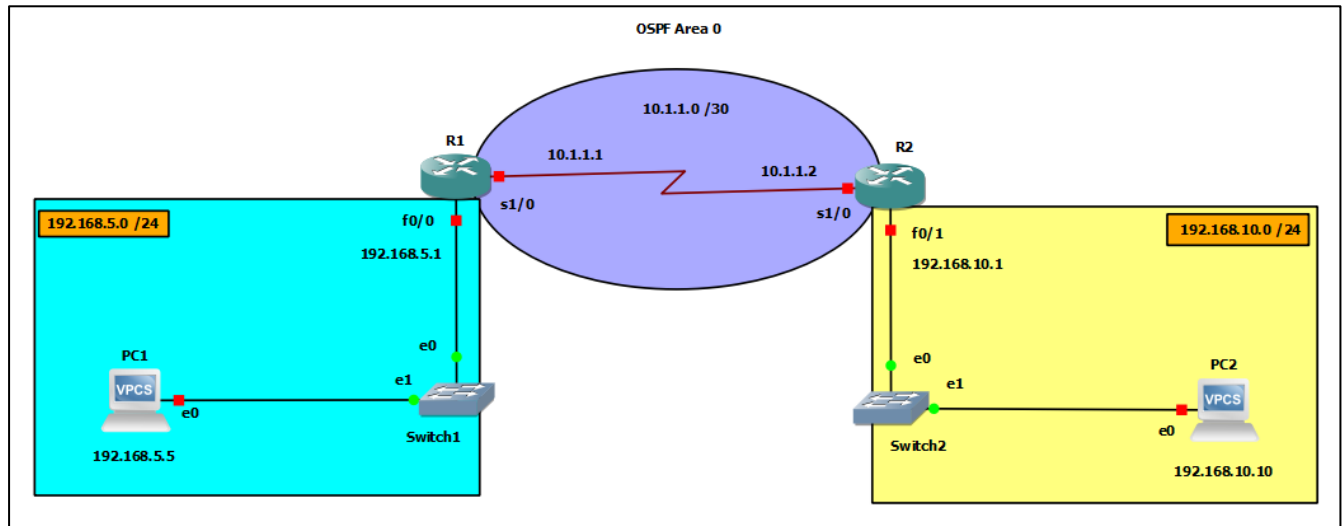
CONCLUSION:

RIPv2 was successfully implemented across the topology. Initially, hosts were unreachable due to the absence of routing information. After enabling RIPv2, both routers exchanged routes and full connectivity between all PCs was achieved. Wireshark verification further confirmed RIPv2 updates and multicast behavior.

PRACTICAL 2B

AIM: Implement the Dynamic Routing Protocol (OSPF)

TOPOLOGY:



STEPS :

STEP 1: Set up Fast Ethernet and Serial Connections on R1

```
R1#conf t
R1(config)#int f0/0
R1(config-if)#ip address 192.168.5.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int s1/0
R1(config-if)#ip address 10.1.1.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#end
```

```
R1(config)#int f0/0
R1(config-if)#ip address 192.168.5.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#
*Mar  1 00:00:51.143: %LINK-3-UPDOWN: Interface FastEthernet0/0,
R1(config)#int s1/0
R1(config-if)#ip address 10.1.1.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#end
R1#
*Mar  1 00:01:15.711: %SYS-5-CONFIG_I: Configured from
```

Type Command: **R1#show ip interface brief** -> To Check interface status quickly

```
R1#show ip interface brief
Interface          IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0    192.168.5.1     YES manual up          up
FastEthernet0/1    unassigned      YES unset  administratively down down
Serial1/0          10.1.1.1        YES manual up          down
Serial1/1          unassigned      YES unset  administratively down down
Serial1/2          unassigned      YES unset  administratively down down
Serial1/3          unassigned      YES unset  administratively down down
```

STEP 2: Set up Fast Ethernet and Serial Connections on R2

```
R2#conf t
R2(config)#int f0/1
R2(config-if)#ip address 192.168.10.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int s1/0
R2(config-if)#ip address 10.1.1.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#end
```

```
R2#conf t
Enter configuration commands, one per line. End with
R2(config)#int f0/1
R2(config-if)#ip address 192.168.10.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
```

```
R2(config)#int s1/0
R2(config-if)#ip address 10.1.1.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#end
R2#
```

Type Command: **R2#show ip interface brief** -> To Check interface status quickly

```
R2#show ip interface brief
Interface          IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0    unassigned      YES unset  administratively down down
FastEthernet0/1    192.168.10.1    YES manual up          up
Serial1/0          10.1.1.2        YES manual up          up
Serial1/1          unassigned      YES unset  administratively down down
Serial1/2          unassigned      YES unset  administratively down down
Serial1/3          unassigned      YES unset  administratively down down
```

STEP 3: Enable OSPF on R1

```
R1#conf t
R1(config)#router ospf 10
R1(config-router)#network 192.168.5.0 0.0.0.255 area 0
R1(config-router)#network 10.1.1.0 0.0.0.3 area 0
R1(config-router)#end
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 10
R1(config-router)#network 192.168.5.0 0.0.0.255 area 0
R1(config-router)#network 10.1.1.0 0.0.0.3 area 0
R1(config-router)#end
R1#
```

```
R2#conf t
R2(config)#router ospf 10
R2(config-router)#network 192.168.10.0 0.0.0.255 area 0
R2(config-router)#network 10.1.1.0 0.0.0.3 area 0
R2(config-router)#
R2(config-router)#end
```

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router ospf 10
R2(config-router)#network 192.168.10.0 0.0.0.255 area 0
R2(config-router)#network 10.1.1.0 0.0.0.3 area 0
R2(config-router)#end
R2#
```

STEP 4: Making sure i.e OSPF doing its job correctly

```
R1#show ip route
```

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external t
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - I
       ia - IS-IS inter area, * - candidate default, U - per-use
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O    192.168.10.0/24 [110/74] via 10.1.1.2, 00:02:10, Serial1/0
C    192.168.5.0/24 is directly connected, FastEthernet0/0
C    10.0.0.0/30 is subnetted, 1 subnets
C      10.1.1.0 is directly connected, Serial1/0
```

R2#show ip route

```
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
       ia - IS-IS inter area, * - candidate default, U - per-us
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.10.0/24 is directly connected, FastEthernet0/1
O    192.168.5.0/24 [110/74] via 10.1.1.1, 00:02:21, Serial1/0
    10.0.0.0/30 is subnetted, 1 subnets
C    10.1.1.0 is directly connected, Serial1/0
```

STEP 5: Manually set an IP address on PC1 and PC2.

PC1> ip 192.168.5.5/24 192.168.5.1

```
PC1 - PuTTY
PC1> ip 192.168.5.5/24 192.168.5.1
Checking for duplicate address...
PC1 : 192.168.5.5 255.255.255.0 gateway 192.168.5.1
```

PC2> ip 192.168.10.10/24 192.168.10.1

```
PC2 - PuTTY
PC2> ip 192.168.10.10/24 192.168.10.1
Checking for duplicate address...
PC1 : 192.168.10.10 255.255.255.0 gateway 192.168.10.1
```

STEP 6: Get the summary of the interfaces on PC1 and PC2.

PC1> show ip

```
PC1 - PuTTY
PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.5.5/24
GATEWAY    : 192.168.5.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10018
RHOST:PORT : 127.0.0.1:10019
MTU       : 1500
```

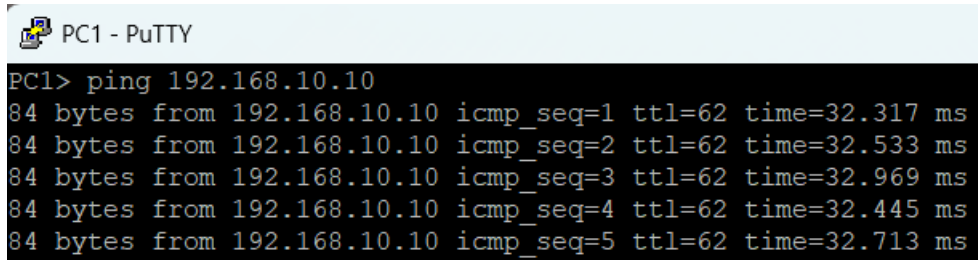
PC2> show ip

```
PC2 - PuTTY
PC2> show ip

NAME       : PC2[1]
IP/MASK    : 192.168.10.10/24
GATEWAY    : 192.168.10.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10020
RHOST:PORT : 127.0.0.1:10021
MTU       : 1500
```

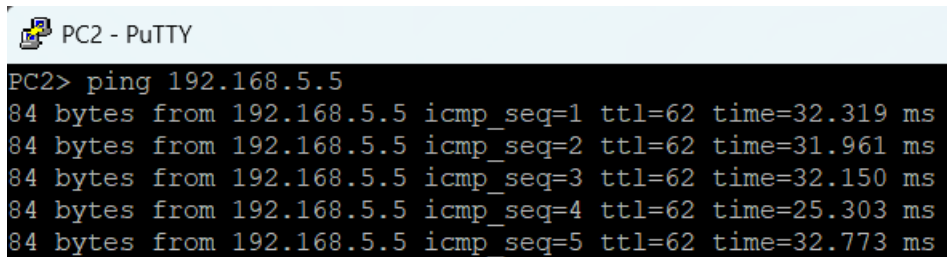
STEP 7: We ping again the interfaces of R1 and R2 on PC1 and PC2 and now they are successful.

PC1> ping 192.168.10.10



```
PC1 - PuTTY
PC1> ping 192.168.10.10
84 bytes from 192.168.10.10 icmp_seq=1 ttl=62 time=32.317 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=62 time=32.533 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=62 time=32.969 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=62 time=32.445 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=62 time=32.713 ms
```

PC2> ping 192.168.5.5



```
PC2 - PuTTY
PC2> ping 192.168.5.5
84 bytes from 192.168.5.5 icmp_seq=1 ttl=62 time=32.319 ms
84 bytes from 192.168.5.5 icmp_seq=2 ttl=62 time=31.961 ms
84 bytes from 192.168.5.5 icmp_seq=3 ttl=62 time=32.150 ms
84 bytes from 192.168.5.5 icmp_seq=4 ttl=62 time=25.303 ms
84 bytes from 192.168.5.5 icmp_seq=5 ttl=62 time=32.773 ms
```

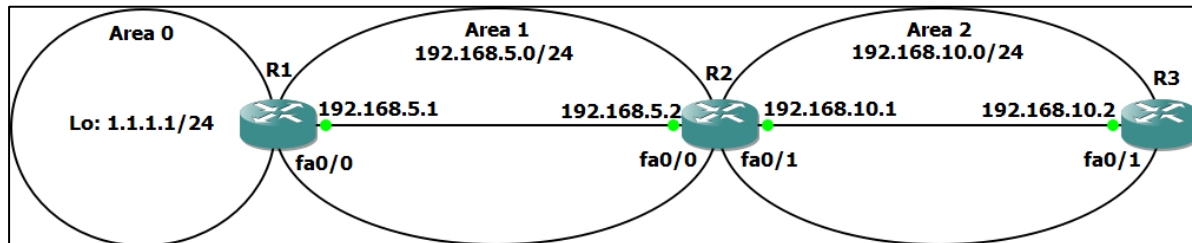
CONCLUSION:

OSPF Area 0 was configured successfully. Both routers formed adjacencies and exchanged routing information. After OSPF convergence, end-to-end connectivity between PCs was verified through successful ping tests, confirming the correct operation of OSPF.

PRACTICAL 3A

AIM: Implementing the concept of OSPF Virtual -Link Configuration

TOPOLOGY:



STEPS:

STEP 1: Configure interfaces on routers R1, R2, R3.

```

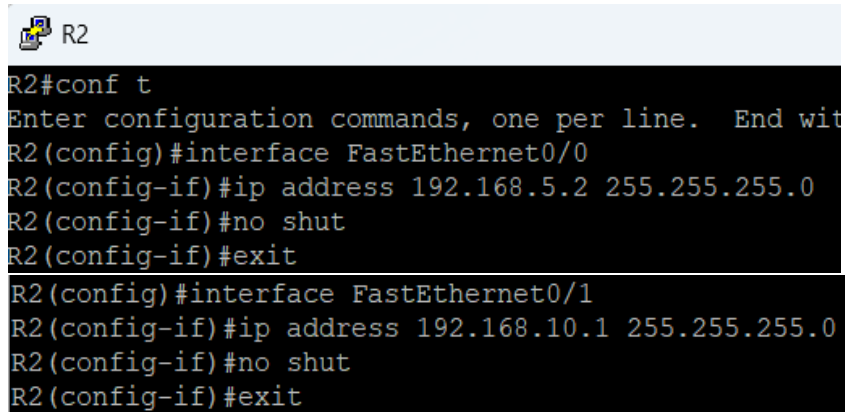
R1#conf t
R1(config)#interface Loopback0
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.5.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
  
```

```

R1
R1#conf t
Enter configuration commands, one per line. End with Ctrl-Z to exit.
R1(config)#interface Loopback0
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.5.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
  
```

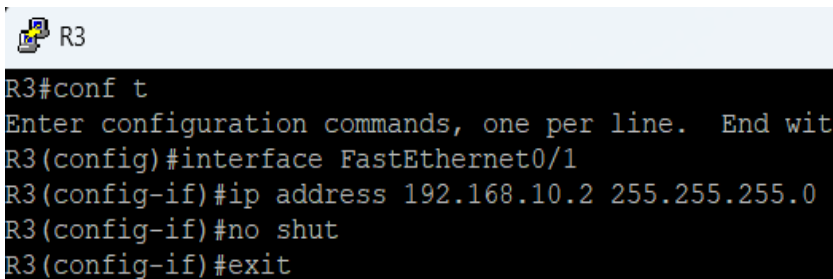


```
R2#conf t
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 192.168.5.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface FastEthernet0/1
R2(config-if)#ip address 192.168.10.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

A screenshot of a terminal window titled 'R2'. The terminal shows the following commands and prompts: R2#conf t, Enter configuration commands, one per line. End with a backslash, R2(config)#interface FastEthernet0/0, R2(config-if)#ip address 192.168.5.2 255.255.255.0, R2(config-if)#no shut, R2(config-if)#exit, R2(config)#interface FastEthernet0/1, R2(config-if)#ip address 192.168.10.1 255.255.255.0, R2(config-if)#no shut, and R2(config-if)#exit.

```
R2#conf t
Enter configuration commands, one per line. End with a backslash
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 192.168.5.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface FastEthernet0/1
R2(config-if)#ip address 192.168.10.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R3#conf t
R3(config)#interface FastEthernet0/1
R3(config-if)#ip address 192.168.10.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

A screenshot of a terminal window titled 'R3'. The terminal shows the following commands and prompts: R3#conf t, Enter configuration commands, one per line. End with a backslash, R3(config)#interface FastEthernet0/1, R3(config-if)#ip address 192.168.10.2 255.255.255.0, R3(config-if)#no shut, and R3(config-if)#exit.

```
R3#conf t
Enter configuration commands, one per line. End with a backslash
R3(config)#interface FastEthernet0/1
R3(config-if)#ip address 192.168.10.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

STEP 2: Execute “show ip route” command on all 3 routers.

R1#show ip route

```

R1
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

1.0.0.0/24 is subnetted, 1 subnets
C       1.1.1.0 is directly connected, Loopback0
C       192.168.5.0/24 is directly connected, FastEthernet0/0

```

R2#show ip route

```

R2
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C       192.168.10.0/24 is directly connected, FastEthernet0/1
C       192.168.5.0/24 is directly connected, FastEthernet0/0

```

R3#show ip route

```

R3
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C       192.168.10.0/24 is directly connected, FastEthernet0/1

```

STEP 3: Use network commands to turn on OSPF on R1, R2, and R3.

```

R1#conf t
R1(config)#router ospf 1
R1(config-router)#network 192.168.5.0 0.0.0.255 area 1
R1(config-router)#network 1.1.1.0 0.0.0.255 area 0

```

```

R1
R1#conf t
Enter configuration commands, one per line. End with C
R1(config)#router ospf 1
R1(config-router)#network 192.168.5.0 0.0.0.255 area 1
R1(config-router)#network 1.1.1.0 0.0.0.255 area 0
R1(config-router)#exit

```

```

R2#conf t
R2(config)#router ospf 1
R2(config-router)#network 192.168.5.0 0.0.0.255 area 1

```

```

R2
R2#conf t
Enter configuration commands, one per line. End with CN
R2(config)#router ospf 1
R2(config-router)#network 192.168.5.0 0.0.0.255 area 1
R2(config-router)#network 192.168.10.0 0.0.0.255 area 2
R2(config-router)#exit

```

```

R3#conf t
R3(config)#router ospf 1
R3(config-router)#network 192.168.10.0 0.0.0.255 area 2
R3(config-router)#exit

```

```

R3
R3#conf t
Enter configuration commands, one per line. End with CN
R3(config)#router ospf 1
R3(config-router)#network 192.168.10.0 0.0.0.255 area 2
R3(config-router)#exit

```

STEP 4: Using the **show ip ospf neighbor** command, we can view the OSPF neighbor that is formed through the loopback interface used for the virtual link.

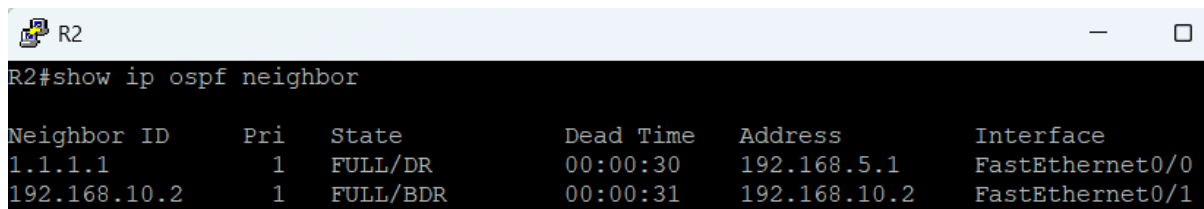
R1#show ip ospf neighbor

```

R1
R1#show ip ospf neighbor

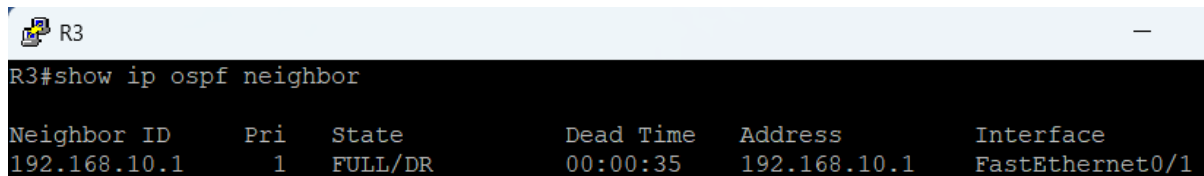
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.10.1	1	FULL/BDR	00:00:34	192.168.5.2	FastEthernet0/0

R2#show ip ospf neighbor


```
R2#show ip ospf neighbor
```

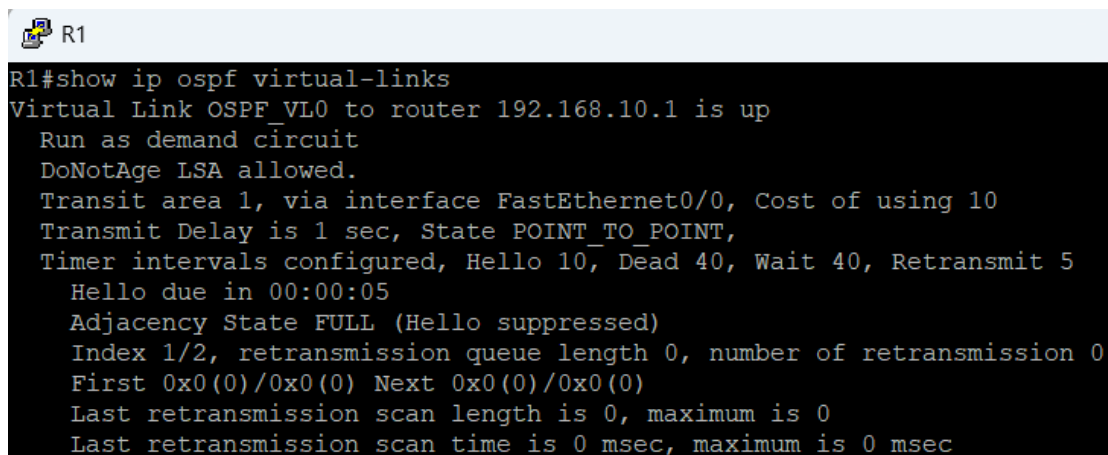
Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/DR	00:00:30	192.168.5.1	FastEthernet0/0
192.168.10.2	1	FULL/BDR	00:00:31	192.168.10.2	FastEthernet0/1

R3#show ip ospf neighbor


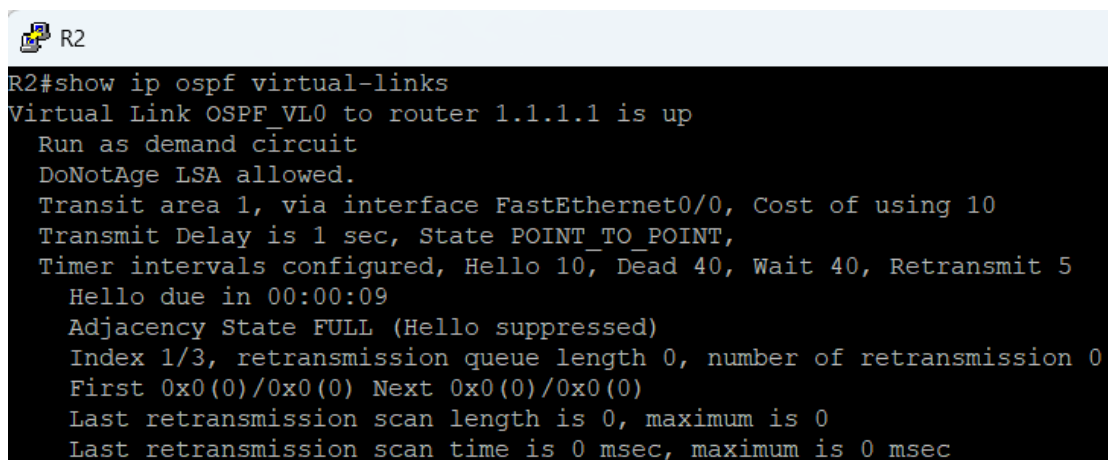
```
R3#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.10.1	1	FULL/DR	00:00:35	192.168.10.1	FastEthernet0/1

STEP 5: The **show ip ospf virtual-links** command lets us verify the virtual link's status, the interface it uses, and other related details.

R1#show ip ospf virtual-links


```
R1#show ip ospf virtual-links
Virtual Link OSPF_VL0 to router 192.168.10.1 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 1, via interface FastEthernet0/0, Cost of using 10
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Adjacency State FULL (Hello suppressed)
  Index 1/2, retransmission queue length 0, number of retransmission 0
  First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
  Last retransmission scan length is 0, maximum is 0
  Last retransmission scan time is 0 msec, maximum is 0 msec
```

R2#show ip ospf virtual-links


```
R2#show ip ospf virtual-links
Virtual Link OSPF_VL0 to router 1.1.1.1 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 1, via interface FastEthernet0/0, Cost of using 10
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:09
  Adjacency State FULL (Hello suppressed)
  Index 1/3, retransmission queue length 0, number of retransmission 0
  First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
  Last retransmission scan length is 0, maximum is 0
  Last retransmission scan time is 0 msec, maximum is 0 msec
```

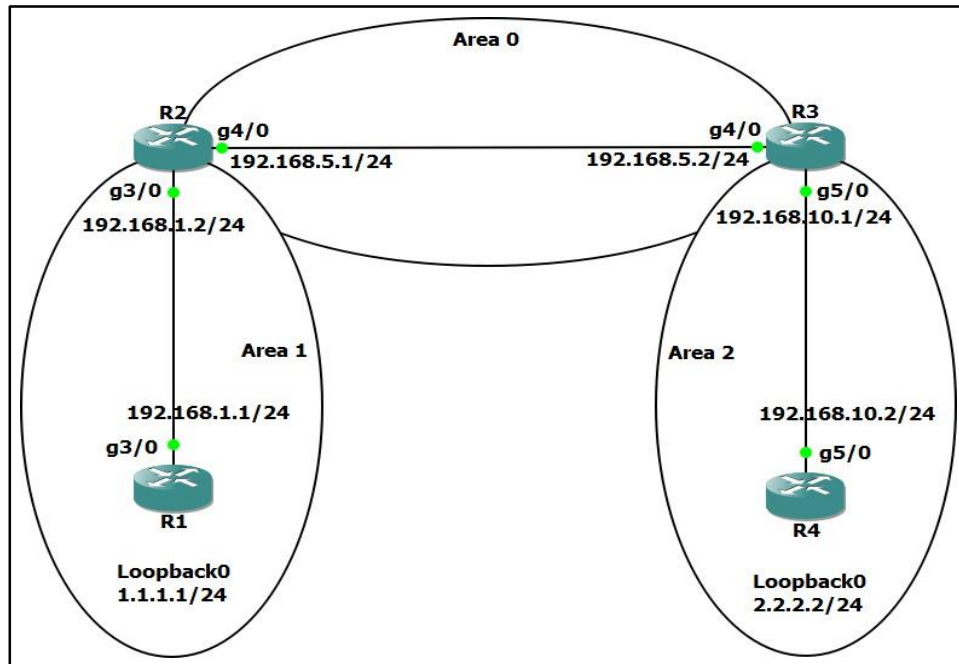
CONCLUSION:

The OSPF Virtual Link was created and verified successfully. Routers established adjacencies through the virtual link, ensuring Area 1's connectivity to the Backbone Area. Commands like **show ip ospf neighbor** and **show ip ospf virtual-links** confirmed proper operation of the virtual link and OSPF routing.

PRACTICAL 3B

AIM: Implementing the concept of OSPF Standard Area and Backbone Area

TOPOLOGY:



STEPS:

STEP 1: Configure interfaces on Routers R1, R2, R3 & R4.

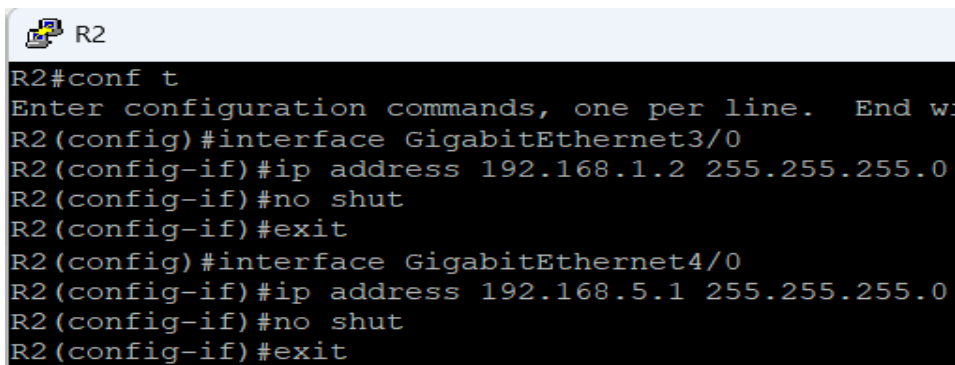
```
R1#conf t
R1(config)#interface Loopback0
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface GigabitEthernet3/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

R1

```
R1#conf t
Enter configuration commands, one per line. End with Ctrl-Z to exit.
R1(config)#interface Loopback0
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

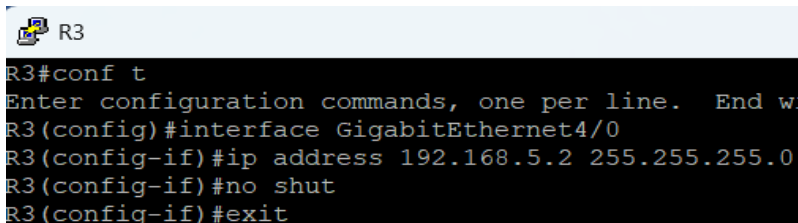
```
R1(config)#interface GigabitEthernet3/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

```
R2#conf t
R2(config)#interface GigabitEthernet3/0
R2(config-if)#ip address 192.168.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface GigabitEthernet4/0
R2(config-if)#ip address 192.168.5.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

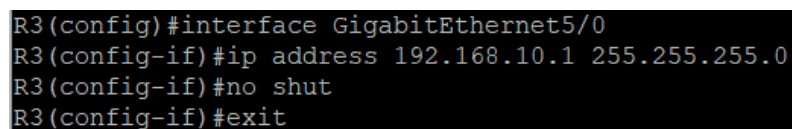
A screenshot of a terminal window titled 'R2' showing the configuration of interfaces GigabitEthernet3/0 and GigabitEthernet4/0. The commands entered are: R2#conf t, R2(config)#interface GigabitEthernet3/0, R2(config-if)#ip address 192.168.1.2 255.255.255.0, R2(config-if)#no shut, R2(config-if)#exit, R2(config)#interface GigabitEthernet4/0, R2(config-if)#ip address 192.168.5.1 255.255.255.0, R2(config-if)#no shut, and R2(config-if)#exit. The prompt 'Enter configuration commands, one per line. End with Ctrl-D' is visible after the first command.

```
R2#conf t
Enter configuration commands, one per line. End with Ctrl-D
R2(config)#interface GigabitEthernet3/0
R2(config-if)#ip address 192.168.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface GigabitEthernet4/0
R2(config-if)#ip address 192.168.5.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R3#conf t
R3(config)#interface GigabitEthernet4/0
R3(config-if)#ip address 192.168.5.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#interface GigabitEthernet5/0
R3(config-if)#ip address 192.168.10.1 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

A screenshot of a terminal window titled 'R3' showing the configuration of interfaces GigabitEthernet4/0 and GigabitEthernet5/0. The commands entered are: R3#conf t, R3(config)#interface GigabitEthernet4/0, R3(config-if)#ip address 192.168.5.2 255.255.255.0, R3(config-if)#no shut, and R3(config-if)#exit. The prompt 'Enter configuration commands, one per line. End with Ctrl-D' is visible after the first command.

```
R3#conf t
Enter configuration commands, one per line. End with Ctrl-D
R3(config)#interface GigabitEthernet4/0
R3(config-if)#ip address 192.168.5.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

A screenshot showing the continuation of the configuration for R3, specifically the configuration of interface GigabitEthernet5/0. The commands entered are: R3(config)#interface GigabitEthernet5/0, R3(config-if)#ip address 192.168.10.1 255.255.255.0, R3(config-if)#no shut, and R3(config-if)#exit.

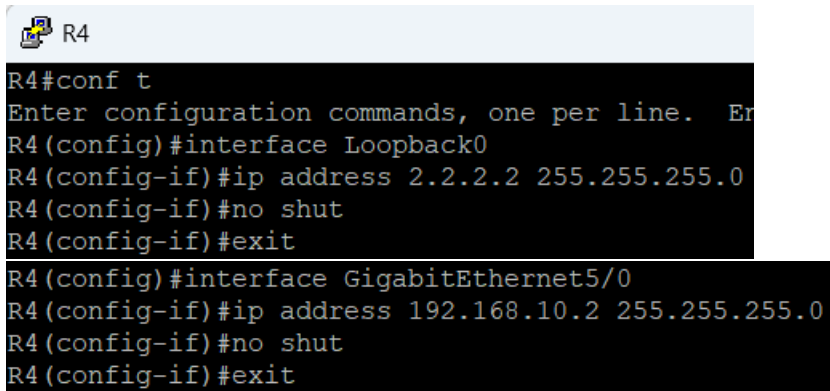
```
R3(config)#interface GigabitEthernet5/0
R3(config-if)#ip address 192.168.10.1 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

```
R4#conf t
R4(config)#interface Loopback0
R4(config-if)#ip address 2.2.2.2 255.255.255.0
R4(config-if)#no shut
```

```

R4(config-if)#exit
R4(config)#interface GigabitEthernet5/0
R4(config-if)#ip address 192.168.10.2 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit

```



```

R4
R4#conf t
Enter configuration commands, one per line.  End with C
R4(config)#interface Loopback0
R4(config-if)#ip address 2.2.2.2 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#interface GigabitEthernet5/0
R4(config-if)#ip address 192.168.10.2 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit

```

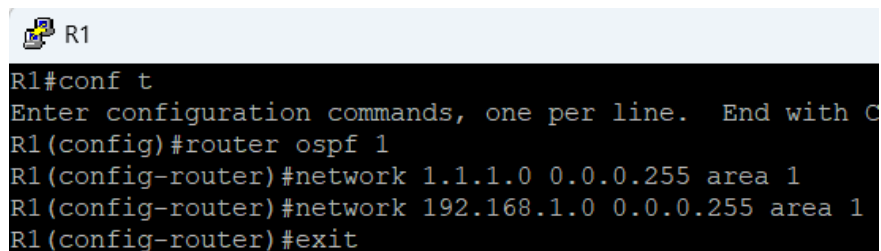
STEP 2: Configure OSPF on all routers as shown below.

- **Configure R1 For Area 1:**

```

R1#conf t
R1(config)#router ospf 1
R1(config-router)#network 1.1.1.10 0.0.0.255 area 1
R1(config-router)#network 192.168.1.0 0.0.0.255 area 1
R1(config-router)#exit

```



```

R1
R1#conf t
Enter configuration commands, one per line.  End with C
R1(config)#router ospf 1
R1(config-router)#network 1.1.1.0 0.0.0.255 area 1
R1(config-router)#network 192.168.1.0 0.0.0.255 area 1
R1(config-router)#exit

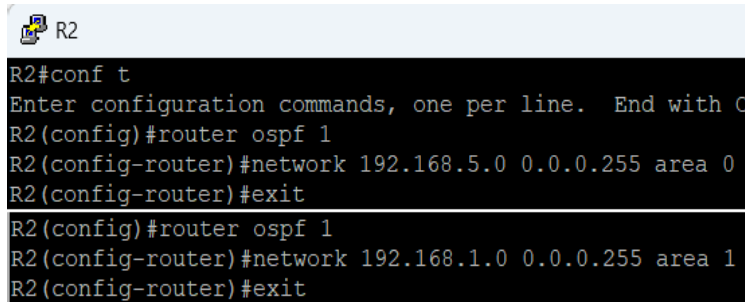
```

- **Configure R2 for Backbone Area (Area 0) and Area 1:**

```

R2#conf t
R2(config)#router ospf 1
R2(config-router)#network 192.168.5.0 0.0.0.255 area 0
R2(config-router)#exit
R2(config)#router ospf 1
R2(config-router)#network 192.168.1.0 0.0.0.255 area 1
R2(config-router)#exit

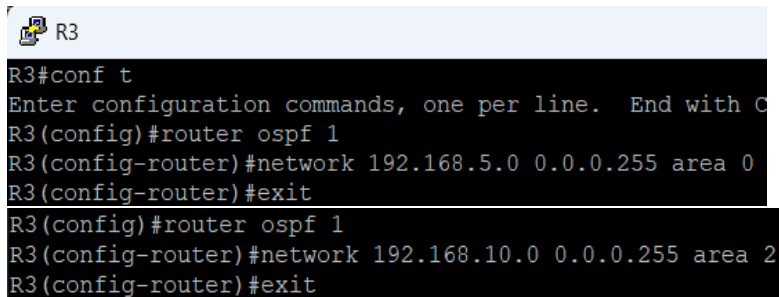
```

```
R2
R2#conf t
Enter configuration commands, one per line.  End with C
R2(config)#router ospf 1
R2(config-router)#network 192.168.5.0 0.0.0.255 area 0
R2(config-router)#exit
R2(config)#router ospf 1
R2(config-router)#network 192.168.1.0 0.0.0.255 area 1
R2(config-router)#exit
```

- **Configure R3 for Backbone Area (Area 0) and Area 2:**

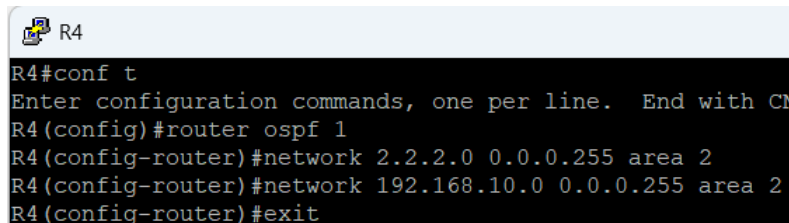
```
R3#conf t
R3(config)#router ospf 1
R3(config-router)#network 192.168.5.0 0.0.0.255 area 0
R3(config-router)#exit
R3(config)#router ospf 1
R3(config-router)#network 192.168.10.0 0.0.0.255 area 2
R3(config-router)#exit
```



```
R3
R3#conf t
Enter configuration commands, one per line.  End with C
R3(config)#router ospf 1
R3(config-router)#network 192.168.5.0 0.0.0.255 area 0
R3(config-router)#exit
R3(config)#router ospf 1
R3(config-router)#network 192.168.10.0 0.0.0.255 area 2
R3(config-router)#exit
```

- **Configure R4 for Area 2:**

```
R4#conf t
R4(config)#router ospf 1
R4(config-router)#network 2.2.2.0 0.0.0.255 area 2
R4(config-router)#network 192.168.10.0 0.0.0.255 area 2
R4(config-router)#exit
```



```
R4
R4#conf t
Enter configuration commands, one per line.  End with CN
R4(config)#router ospf 1
R4(config-router)#network 2.2.2.0 0.0.0.255 area 2
R4(config-router)#network 192.168.10.0 0.0.0.255 area 2
R4(config-router)#exit
```

STEP 3: Verify the network.

R1#show ip ospf neighbor

 R1


```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.5.1	1	FULL/BDR	00:00:35	192.168.1.2	GigabitEthernet3/0

R1 has formed a neighbor adjacency with R2.

R2#show ip ospf neighbor

 R2


```
R2#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.10.1	1	FULL/DR	00:00:33	192.168.5.2	GigabitEthernet4/0
1.1.1.1	1	FULL/DR	00:00:32	192.168.1.1	GigabitEthernet3/0

R2 has formed neighbor adjacencies with R1 and R3. The “show ip ospf neighbor” command, however, it does not tell us anything about the areas that are used, so we check it using the “show ip ospf neighbor detail” command.

R2#show ip ospf neighbour detail

 R2


```
R2#show ip ospf neighbor detail
```

```
Neighbor 192.168.10.1, interface address 192.168.5.2
  In the area 0 via interface GigabitEthernet4/0
  Neighbor priority is 1, State is FULL, 6 state changes
  DR is 192.168.5.2 BDR is 192.168.5.1
  Options is 0x12 in Hello (E-bit, L-bit)
  Options is 0x52 in DBD (E-bit, L-bit, O-bit)
  LLS Options is 0x1 (LR)
  Dead timer due in 00:00:32
  Neighbor is up for 01:05:50
  Index 1/1, retransmission queue length 0, number of retransmission 0
  First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
  Last retransmission scan length is 0, maximum is 0
  Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 1.1.1.1, interface address 192.168.1.1
  In the area 1 via interface GigabitEthernet3/0
  Neighbor priority is 1, State is FULL, 6 state changes
  DR is 192.168.1.1 BDR is 192.168.1.2
  Options is 0x12 in Hello (E-bit, L-bit)
  Options is 0x52 in DBD (E-bit, L-bit, O-bit)
  LLS Options is 0x1 (LR)
  Dead timer due in 00:00:36
  Neighbor is up for 01:02:41
  Index 1/2, retransmission queue length 0, number of retransmission 0
  First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
  Last retransmission scan length is 0, maximum is 0
  Last retransmission scan time is 0 msec, maximum is 0 msec
```

R3#show ip ospf neighbor

 R3


```
R3#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.5.1	1	FULL/BDR	00:00:31	192.168.5.1	GigabitEthernet4/0
2.2.2.2	1	FULL/BDR	00:00:37	192.168.10.2	GigabitEthernet5/0

R3 has formed neighbor adjacencies with R2 and R4. The “show ip ospf neighbor” command, however, it does not tell us anything about the areas that are used, so we check it using the “show ip ospf neighbor detail” command.

R3#show ip ospf neighbour detail

```

R3
R3#show ip ospf neighbor detail
Neighbor 192.168.5.1, interface address 192.168.5.1
  In the area 0 via interface GigabitEthernet4/0
  Neighbor priority is 1, State is FULL, 6 state changes
  DR is 192.168.5.2 BDR is 192.168.5.1
  Options is 0x12 in Hello (E-bit, L-bit)
  Options is 0x52 in DBD (E-bit, L-bit, O-bit)
  LLS Options is 0x1 (LR)
  Dead timer due in 00:00:32
  Neighbor is up for 00:25:39
  Index 1/1, retransmission queue length 0, number of retransmission 0
  First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
  Last retransmission scan length is 0, maximum is 0
  Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 2.2.2.2, interface address 192.168.10.2
  In the area 2 via interface GigabitEthernet5/0
  Neighbor priority is 1, State is FULL, 6 state changes
  DR is 192.168.10.1 BDR is 192.168.10.2
  Options is 0x12 in Hello (E-bit, L-bit)
  Options is 0x52 in DBD (E-bit, L-bit, O-bit)
  LLS Options is 0x1 (LR)
  Dead timer due in 00:00:38
  Neighbor is up for 00:20:48
  Index 1/2, retransmission queue length 0, number of retransmission 0
  First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
  Last retransmission scan length is 0, maximum is 0
  Last retransmission scan time is 0 msec, maximum is 0 msec

```

Another good command to find area information is show ip protocols:

R2#show ip protocols

```

R2
R2#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.5.1
  It is an area border router
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.1.0 0.0.0.255 area 1
    192.168.5.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:27:38
    192.168.10.1     110          00:26:00
  Distance: (default is 110)

```

R3#show ip protocols

```

R3
R3#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.10.1
  It is an area border router
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.5.0 0.0.0.255 area 0
    192.168.10.0 0.0.0.255 area 2
  Routing Information Sources:
    Gateway         Distance      Last Update
    2.2.2.2          110          00:25:51
    192.168.5.1      110          00:26:42
  Distance: (default is 110)

```

STEP 4: Check the routing tables for all the routers.

R1#show ip route ospf

```

R1
R1#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    2.0.0.0/32 is subnetted, 1 subnets
O IA   2.2.2.2 [110/4] via 192.168.1.2, 00:26:25, GigabitEthernet3/0
O IA   192.168.5.0/24 [110/2] via 192.168.1.2, 00:28:08, GigabitEthernet3/0
O IA   192.168.10.0/24 [110/3] via 192.168.1.2, 00:27:07, GigabitEthernet3/0

```

R2#show ip route ospf

```

R2
R2#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/2] via 192.168.1.1, 00:28:44, GigabitEthernet3/0
    2.0.0.0/32 is subnetted, 1 subnets
O IA    2.2.2.2 [110/3] via 192.168.5.2, 00:27:06, GigabitEthernet4/0
O IA    192.168.10.0/24 [110/2] via 192.168.5.2, 00:27:47, GigabitEthernet4/0

```

R3#show ip route ospf

```

R3
R3#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O IA   1.1.1.1 [110/3] via 192.168.5.1, 00:28:12, GigabitEthernet4/0
      2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/2] via 192.168.10.2, 00:27:20, GigabitEthernet5/0
O IA   192.168.1.0/24 [110/2] via 192.168.5.1, 00:28:12, GigabitEthernet4/0

```

R4# show ip route ospf

```

R4
R4#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O IA   1.1.1.1 [110/4] via 192.168.10.1, 00:27:31, GigabitEthernet5/0
O IA   192.168.1.0/24 [110/3] via 192.168.10.1, 00:27:31, GigabitEthernet5/0
O IA   192.168.5.0/24 [110/2] via 192.168.10.1, 00:27:31, GigabitEthernet5/0

```

STEP 5:

A test ping is performed between R1 and R4 to verify the successful operation of our multi-area OSPF configuration.

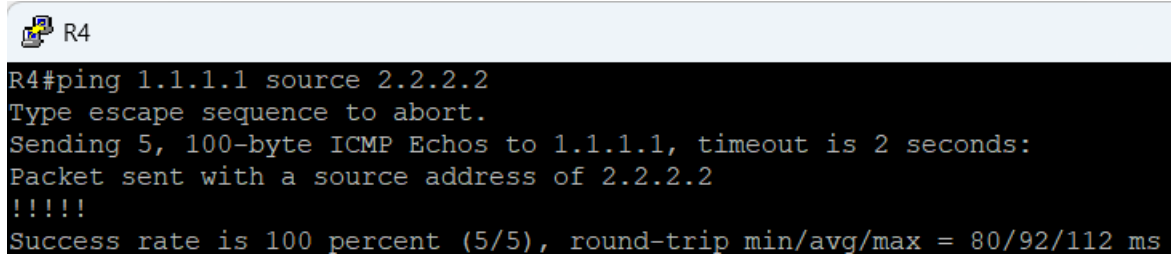
R1#ping 2.2.2.2 source 1.1.1.1

```

R1
R1#ping 2.2.2.2 source 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
Packet sent with a source address of 1.1.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/92/100 ms

```

R4#ping 1.1.1.1 source 2.2.2.2

A screenshot of a terminal window with a light blue title bar containing a router icon and the text 'R4'. The terminal has a black background with green text. The output of the command 'R4#ping 1.1.1.1 source 2.2.2.2' is displayed, showing the command execution, escape sequence instructions, packet sending details, five successful pings represented by exclamation marks, and the final success rate and round-trip times.

```
R4#ping 1.1.1.1 source 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
Packet sent with a source address of 2.2.2.2
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/92/112 ms
```

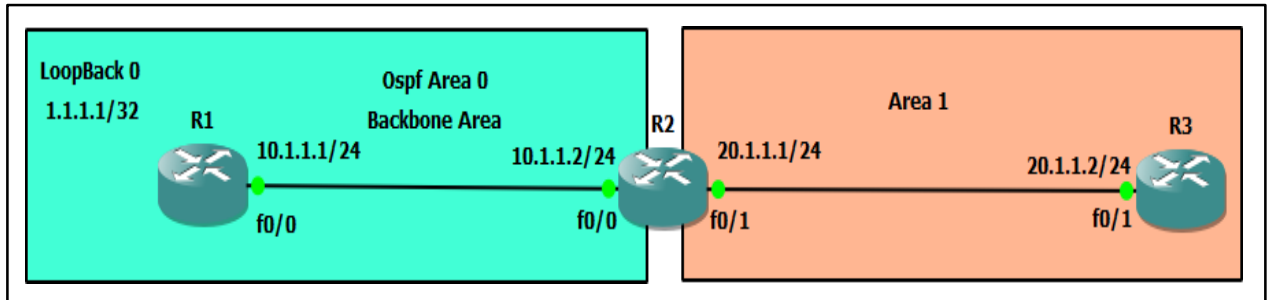
CONCLUSION:

Multi-area OSPF was configured successfully across all routers. Adjacencies formed properly within and between areas, and OSPF learned routes were visible in the routing tables. End-to-end ping tests between R1 and R4 confirmed successful multi-area routing and correct design of backbone and non-backbone areas.

PRACTICAL 3C

AIM: Implementing the concept of OSPF Stub and Totally Stubby Area

TOPOLOGY:



STEPS

STEP 1: Configure interfaces on routers R1, R2 and R3.

```

R1#conf t
R1(config)#int f0/0
R1(config-if)#ip address 10.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int loopback0
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#no shut
R1(config-if)#exit
  
```

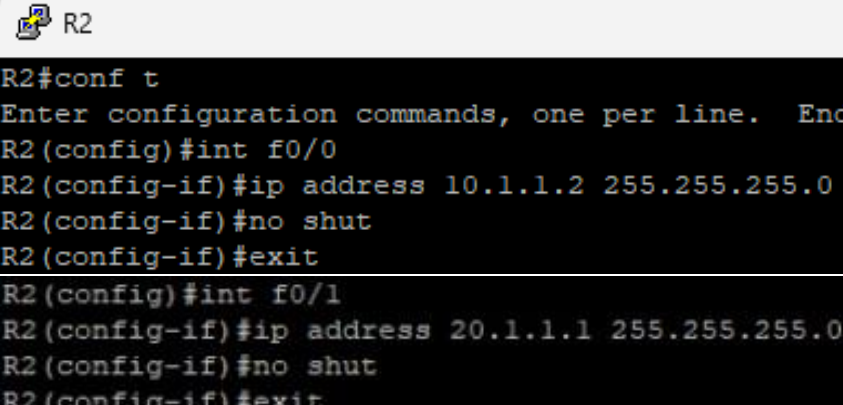
```

R1
R1#conf t
Enter configuration commands, one per line. End
R1(config)#int f0/0

R1(config-if)#ip address 10.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit

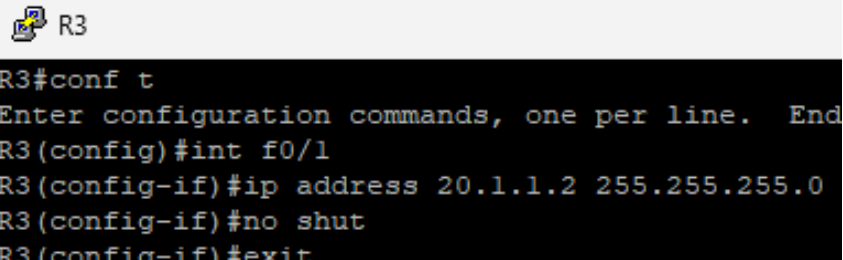
R1(config)#int loopback 0
R1(config-if)#
*Mar  1 00:04:46.063: %LINEPROTO-5-UPDOWN: Line p
changed state to up
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#no shut
R1(config-if)#exit
  
```

```
R2#conf t
R2(config)#int f0/0
R2(config-if)#ip address 10.1.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int f0/1
R2(config-if)#ip address 20.1.1.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

A screenshot of a terminal window titled 'R2' showing the configuration of interfaces f0/0 and f0/1. The commands are entered line by line, and the prompt changes from R2# to R2(config) and then to R2(config-if) for each interface configuration. The output shows the successful assignment of IP addresses and the disabling of shutdown on both interfaces.

```
R2
R2#conf t
Enter configuration commands, one per line.  End
R2(config)#int f0/0
R2(config-if)#ip address 10.1.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int f0/1
R2(config-if)#ip address 20.1.1.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R3#conf t
R3(config)#int f0/1
R3(config-if)#ip address 20.1.1.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

A screenshot of a terminal window titled 'R3' showing the configuration of interface f0/1. The commands are entered line by line, and the prompt changes from R3# to R3(config) and then to R3(config-if) for the interface configuration. The output shows the successful assignment of IP address and the disabling of shutdown on interface f0/1.

```
R3
R3#conf t
Enter configuration commands, one per line.  End
R3(config)#int f0/1
R3(config-if)#ip address 20.1.1.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

STEP 2: Set up the OSPF process on all three routers using the **router ospf <process-id>** command.

```
R1(config)#router ospf 1
R1(config-route)#exit
```


 R1

```
R1(config)#router ospf 1
R1(config-router)#exit
```

```
R2(config)#router ospf 1
R2(config-router)#exit
```

 R2

```
R2(config)#router ospf 1
R2(config-router)#exit
```

```
R3(config)#router ospf 1
R3(config-router)#exit
```

 R3

```
R3(config)#router ospf 1
R3(config-router)#exit
```

STEP 3: Enable OSPF on the interfaces using the command `ip ospf <process-id> area <area-id>` in interface configuration mode.

```
R1(config)#interface loopback0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#int f0/0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#no shut
```

 R1

```
R1(config)#interface loopback0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#int f0/0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#no shut
```

```
R2(config)#int f0/0
R2(config-if)#ip ospf 1 area 0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int f0/1
R2(config-if)#ip ospf 1 area 0
R2(config-if)#no shut
```

```

R2
R2(config)#int f0/0
R2(config-if)#ip ospf 1 area 0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int f0/1
R2(config-if)#ip ospf 1 area 1
R2(config-if)#no shut
R3(config)#int f0/1
R3(config-if)#ip ospf 1 area 1
R3(config-if)#no shut

```

```

R3
R3(config)#int f0/1
R3(config-if)#ip ospf 1 area 1
R3(config-if)#no shut

```

STEP 4: Configuring Area 2 as a Stub Area with the router sub-command “area <area- id> stub”.

```

R2(config)#router ospf 1
R2(config-router)#area 1 stub

```

```

R3(config)#router ospf 1
R3(config-router)#area 1 stub

```

```

R2
R2(config)#router ospf 1
R2(config-router)#area 1 stub

```

```

R3
R3(config)#router ospf 1
R3(config-router)#area 1 stub

```

STEP 5: Verifying OSPF configuration on all the routers.

R1#show ip protocol

```

R1
R1#show ip protocol
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Routing on Interfaces Configured Explicitly (Area 0):
    Loopback0
    FastEthernet0/0
  Reference bandwidth unit is 100 mbps
  Routing Information Sources:
    Gateway         Distance         Last Update
    20.1.1.1         110              00:06:41
  Distance: (default is 110)

```

R2#show ip protocol

```

R2
R2#show ip protocol
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 20.1.1.1
  It is an area border router
  Number of areas in this router is 2. 1 normal 1 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    Routing on Interfaces Configured Explicitly (Area 0):
      FastEthernet0/0
    Routing on Interfaces Configured Explicitly (Area 1):
      FastEthernet0/1
  Reference bandwidth unit is 100 mbps
  Routing Information Sources:
    Gateway         Distance         Last Update
    1.1.1.1          110              00:00:39
  Distance: (default is 110)

```

R3#show ip protocol

```

R3
R3#show ip protocol
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 20.1.1.2
  Number of areas in this router is 1. 0 normal 1 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    Routing on Interfaces Configured Explicitly (Area 1):
      FastEthernet0/1
  Reference bandwidth unit is 100 mbps
  Routing Information Sources:
    Gateway         Distance         Last Update
    20.1.1.1          110              00:00:43
  Distance: (default is 110)

```

STEP 6: Checking the IPv4 routing table on the routers and seeing OSPF route entry.

R1#show ip route

```

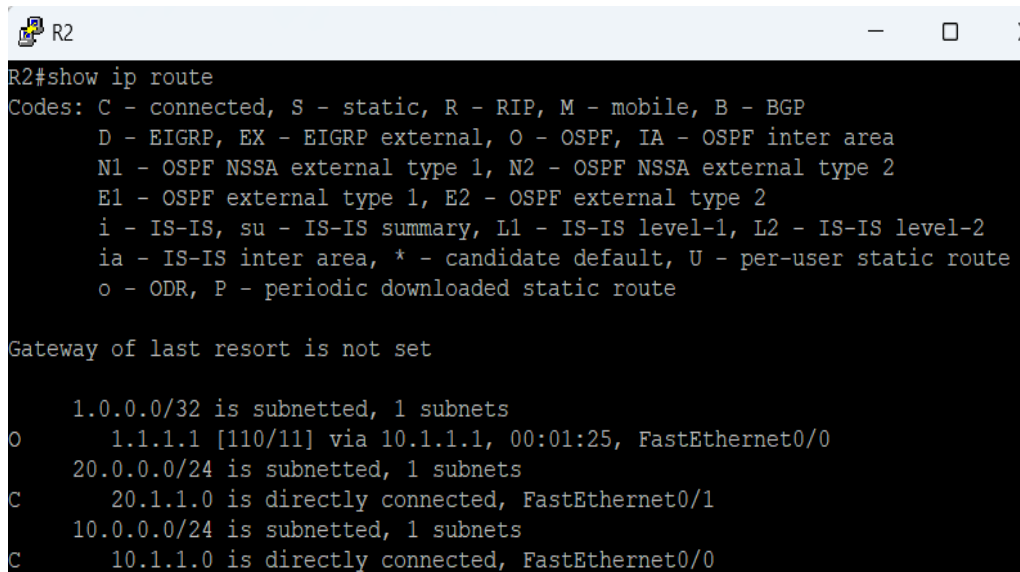
R1
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  1.0.0.0/30 is subnetted, 1 subnets
C      1.1.1.0 is directly connected, Loopback0
  20.0.0.0/24 is subnetted, 1 subnets
O IA   20.1.1.0 [110/20] via 10.1.1.2, 00:07:30, FastEthernet0/0
  10.0.0.0/24 is subnetted, 1 subnets
C      10.1.1.0 is directly connected, FastEthernet0/0

```

R2#show ip route



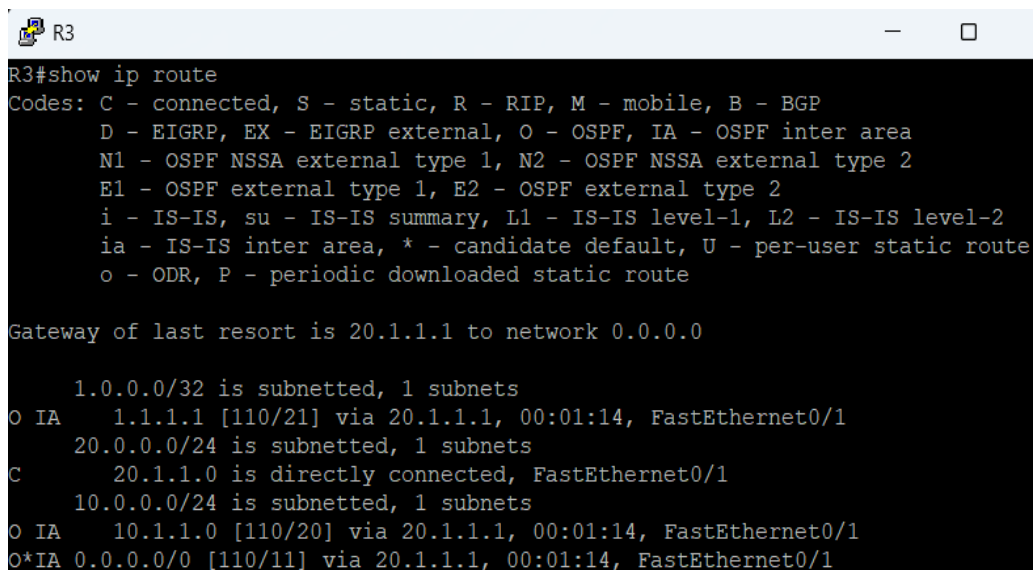
```

R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    1.0.0.0/32 is subnetted, 1 subnets
O      1.1.1.1 [110/11] via 10.1.1.1, 00:01:25, FastEthernet0/0
    20.0.0.0/24 is subnetted, 1 subnets
C      20.1.1.0 is directly connected, FastEthernet0/1
    10.0.0.0/24 is subnetted, 1 subnets
C      10.1.1.0 is directly connected, FastEthernet0/0
  
```

R3#show ip route



```

R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 20.1.1.1 to network 0.0.0.0

    1.0.0.0/32 is subnetted, 1 subnets
O IA   1.1.1.1 [110/21] via 20.1.1.1, 00:01:14, FastEthernet0/1
    20.0.0.0/24 is subnetted, 1 subnets
C      20.1.1.0 is directly connected, FastEthernet0/1
    10.0.0.0/24 is subnetted, 1 subnets
O IA   10.1.1.0 [110/20] via 20.1.1.1, 00:01:14, FastEthernet0/1
O*IA  0.0.0.0/0 [110/11] via 20.1.1.1, 00:01:14, FastEthernet0/1
  
```

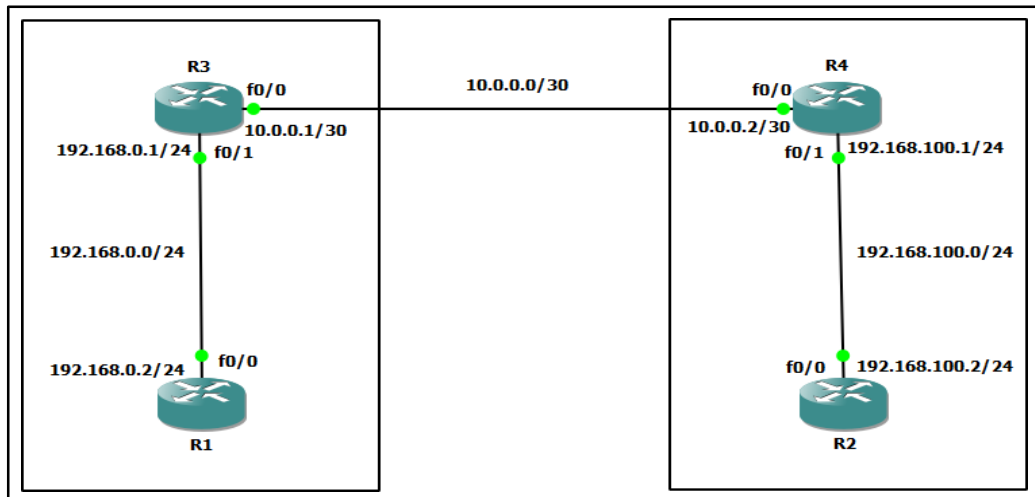
CONCLUSION:

Stub area configuration was implemented correctly. Area 1 was converted into a stub area, and all routers exchanged routing information accordingly. The routing tables reflected default routes as expected in a stub area, and connectivity tests verified that the OSPF stub configuration functioned properly.

PRACTICAL 4A

AIM: Implementing the concept of BGP Protocol

TOPOLOGY:



STEPS

STEP 1: Configure interfaces on routers R1, R2, R3 and R4.

R1#conf t

R1(config)#interface FastEthernet0/0

R1(config-if)#ip address 192.168.0.2 255.255.255.0

R1(config-if)#no shut

R1(config-if)#exit

```

R1
R1#conf t
Enter configuration commands, one per line.  End with Ctrl-Z to exit.
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.0.2 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
  
```

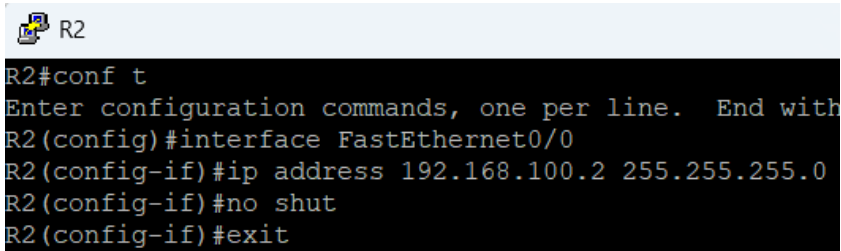
R2#conf t

R2(config)#interface FastEthernet0/0

R2(config-if)#ip address 192.168.100.2 255.255.255.0

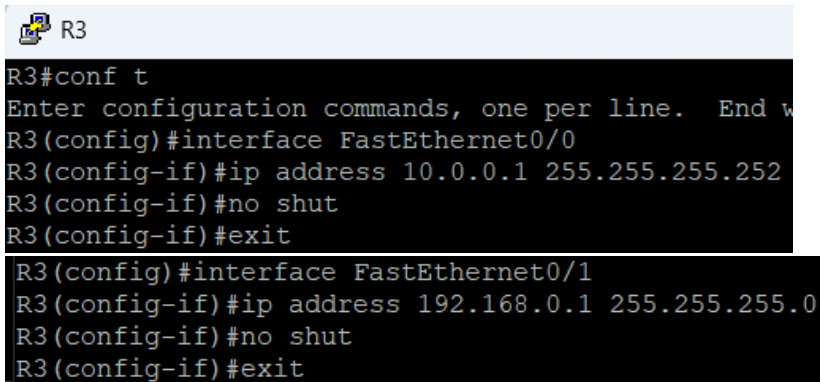
R2(config-if)#no shut

R2(config-if)#exit

A screenshot of a terminal window for router R2. The title bar shows a router icon and the text 'R2'. The terminal text shows the configuration process: entering configuration mode, configuring interface FastEthernet0/0 with IP 192.168.100.2 and mask 255.255.255.0, and exiting configuration mode.

```
R2#conf t
Enter configuration commands, one per line.  End with
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 192.168.100.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R3#conf t
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 10.0.0.1 255.255.255.252
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#interface FastEthernet0/1
R3(config-if)#ip address 192.168.0.1 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

A screenshot of a terminal window for router R3. The title bar shows a router icon and the text 'R3'. The terminal text shows the configuration process: entering configuration mode, configuring interface FastEthernet0/0 with IP 10.0.0.1 and mask 255.255.255.252, and exiting configuration mode. Then, configuring interface FastEthernet0/1 with IP 192.168.0.1 and mask 255.255.255.0, and exiting configuration mode.

```
R3#conf t
Enter configuration commands, one per line.  End w
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 10.0.0.1 255.255.255.252
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#interface FastEthernet0/1
R3(config-if)#ip address 192.168.0.1 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

```
R4#conf t
R4(config)#interface FastEthernet0/0
R4(config-if)#ip address 10.0.0.2 255.255.255.252
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#interface FastEthernet0/1
R4(config-if)#ip address 192.168.100.1 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
```

 R4

```
R4#conf t
Enter configuration commands, one per line.  End with CNTL-Z
R4(config)#interface FastEthernet0/0
R4(config-if)#ip address 10.0.0.2 255.255.255.252
R4(config-if)#no shut
R4(config-if)#exit
```

```
R4(config)#interface FastEthernet0/1
R4(config-if)#ip address 192.168.100.1 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
```

STEP 2: After we configure IP addresses in the interfaces, we configure BGP. The configuration of BGP starts with “router bgp AS” which enters us to the BGP configuration mode. The next step is to configure bgp neighbors with “neighbor (ip- address | peer-group-name) remote-as AS” command. We set neighbors based on ip address. We use the command “network (network-number) mask (network-mask) to tell the router what to advertise.

```
R1#conf t
R1(config)#router bgp 65300
R1(config-router)#neighbor 192.168.0.1 remote-as 65300
R1(config-router)#exit
```

 R1

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL-Z
R1(config)#router bgp 65300
R1(config-router)#neighbor 192.168.0.1 remote-as 65300
R1(config-router)#exit
```

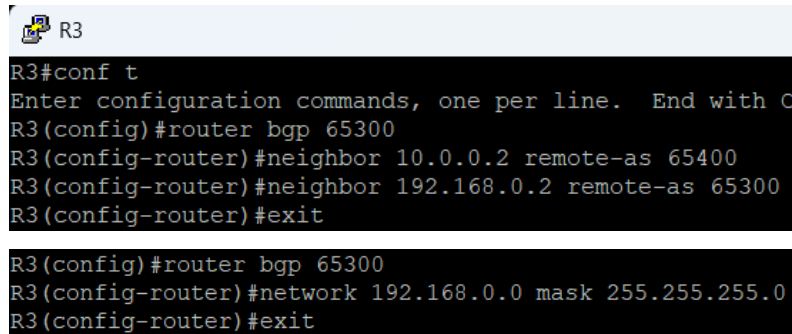
```
R2#conf t
R2(config)#router bgp 65400
R2(config-router)#neighbor 192.168.100.1 remote-as 65400
R2(config-router)#exit
```

 R2

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL-Z
R2(config)#router bgp 65400
R2(config-router)#neighbor 192.168.100.1 remote-as 65400
R2(config-router)#exit
```

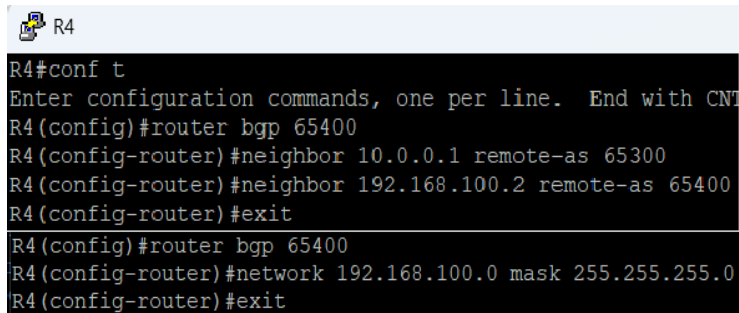
```
R3#conf t
R3(config)#router bgp 65300
```

```
R3(config-router)#neighbor 10.0.0.2 remote-as 65400
R3(config-router)#neighbor 192.168.0.2 remote-as 65300
R3(config-router)#exit
R3(config)#router bgp 65300
R3(config-router)#network 192.168.0.0 mask 255.255.255.0
R3(config-router)#exit
```

A screenshot of a terminal window titled 'R3'. It shows the configuration of router R3. The commands entered are: 'R3#conf t', 'R3(config)#router bgp 65300', 'R3(config-router)#neighbor 10.0.0.2 remote-as 65400', 'R3(config-router)#neighbor 192.168.0.2 remote-as 65300', 'R3(config-router)#exit', 'R3(config)#router bgp 65300', 'R3(config-router)#network 192.168.0.0 mask 255.255.255.0', and 'R3(config-router)#exit'.

```
R3#conf t
Enter configuration commands, one per line. End with CNTL-Z
R3(config)#router bgp 65300
R3(config-router)#neighbor 10.0.0.2 remote-as 65400
R3(config-router)#neighbor 192.168.0.2 remote-as 65300
R3(config-router)#exit
R3(config)#router bgp 65300
R3(config-router)#network 192.168.0.0 mask 255.255.255.0
R3(config-router)#exit
```

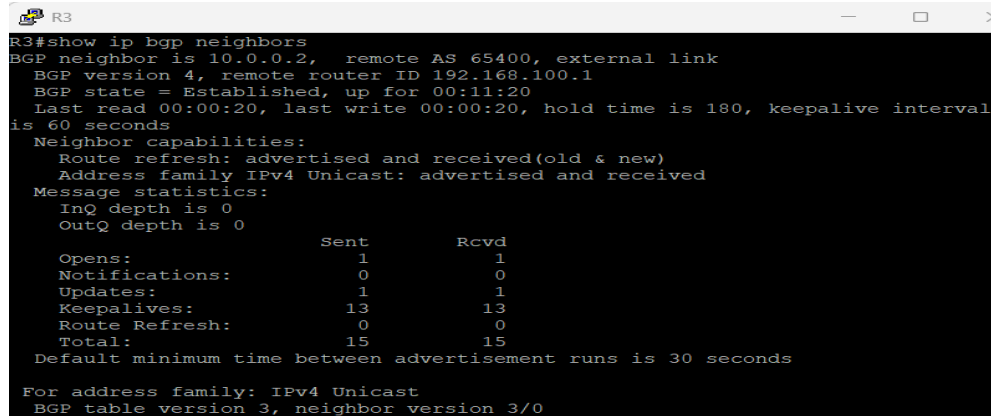
```
R4#conf t
R4(config)#router bgp 65400
R4(config-router)#neighbor 10.0.0.1 remote-as 65300
R4(config-router)#neighbor 192.168.100.2 remote-as 65400
R4(config-router)#exit
R4(config)#router bgp 65400
R4(config-router)#network 192.168.100.0 mask 255.255.255.0
R4(config-router)#exit
```

A screenshot of a terminal window titled 'R4'. It shows the configuration of router R4. The commands entered are: 'R4#conf t', 'R4(config)#router bgp 65400', 'R4(config-router)#neighbor 10.0.0.1 remote-as 65300', 'R4(config-router)#neighbor 192.168.100.2 remote-as 65400', 'R4(config-router)#exit', 'R4(config)#router bgp 65400', 'R4(config-router)#network 192.168.100.0 mask 255.255.255.0', and 'R4(config-router)#exit'.

```
R4#conf t
Enter configuration commands, one per line. End with CNTL-Z
R4(config)#router bgp 65400
R4(config-router)#neighbor 10.0.0.1 remote-as 65300
R4(config-router)#neighbor 192.168.100.2 remote-as 65400
R4(config-router)#exit
R4(config)#router bgp 65400
R4(config-router)#network 192.168.100.0 mask 255.255.255.0
R4(config-router)#exit
```

STEP 3: Showing neighbors on R3.

R3#show ip bgp neighbors



```

R3#show ip bgp neighbors
BGP neighbor is 10.0.0.2, remote AS 65400, external link
  BGP version 4, remote router ID 192.168.100.1
  BGP state = Established, up for 00:11:20
  Last read 00:00:20, last write 00:00:20, hold time is 180, keepalive interval
  is 60 seconds
  Neighbor capabilities:
    Route refresh: advertised and received(old & new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

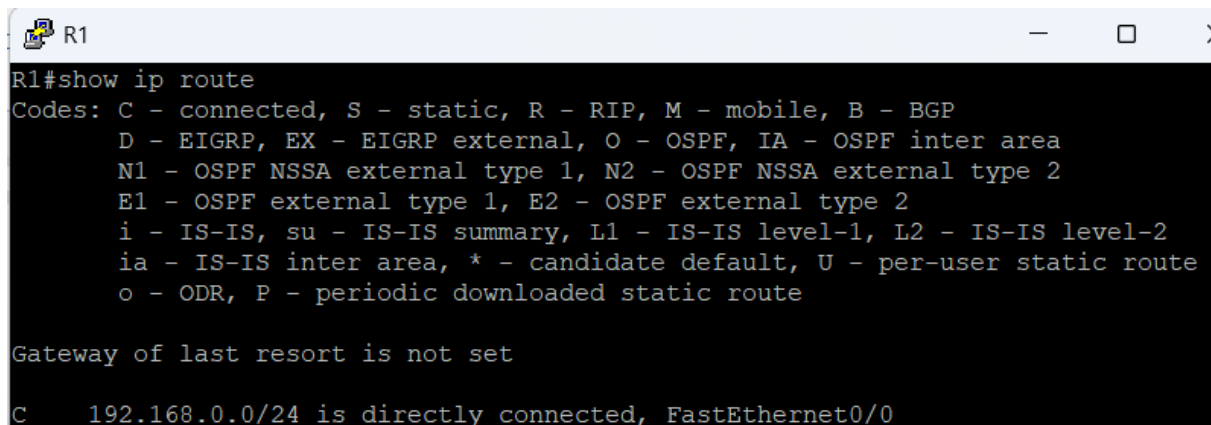
      Sent      Rcvd
Opens:         1         1
Notifications: 0         0
Updates:       1         1
Keepalives:    13        13
Route Refresh: 0         0
Total:         15        15
Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast
BGP table version 3, neighbor version 3/0

```

Step 4: We execute the “show ip route” command on R1.

R1#show ip route



```

R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C       192.168.0.0/24 is directly connected, FastEthernet0/0

```

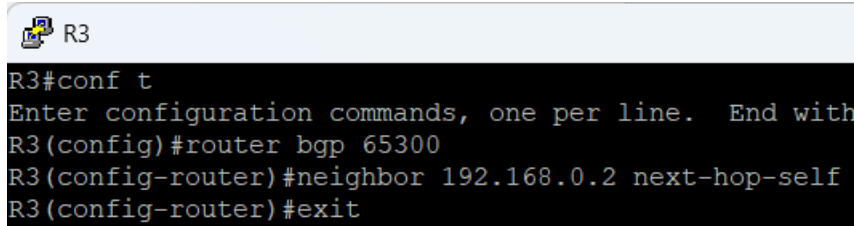
STEP 5: As we can see on R1 we do not have an external route to network 192.168.100.0/24. This is because on R4, which is an external router, routing updates sent, have address 10.0.0.2 as source ip address, but in this case R1 does not have any idea how to get 10.0.0.0 network. BGP on R1 does not put in routing table, a route to 192.168.100.0/24. To overcome this, we have to add on router R3 this command: “neighbor 192.168.0.2 next-hop-self”. This command will change source ip address of routing updates sent to router R1 (will change to its e0/1 ip address). We need to put the command inside router bgp AS configuration.

R3#conf t

R3(config)#router bgp 65300

R3(config-router)#neighbor 192.168.0.2 next-hop-self

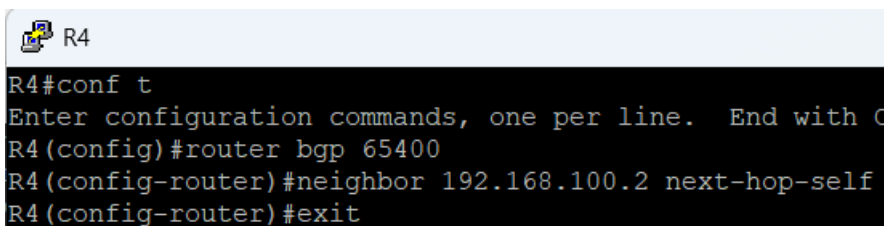
R3(config-router)#exit



```
R3
R3#conf t
Enter configuration commands, one per line. End with
R3(config)#router bgp 65300
R3(config-router)#neighbor 192.168.0.2 next-hop-self
R3(config-router)#exit
```

STEP 6: We need to do the same to R4 to change update source to neighbor R1.

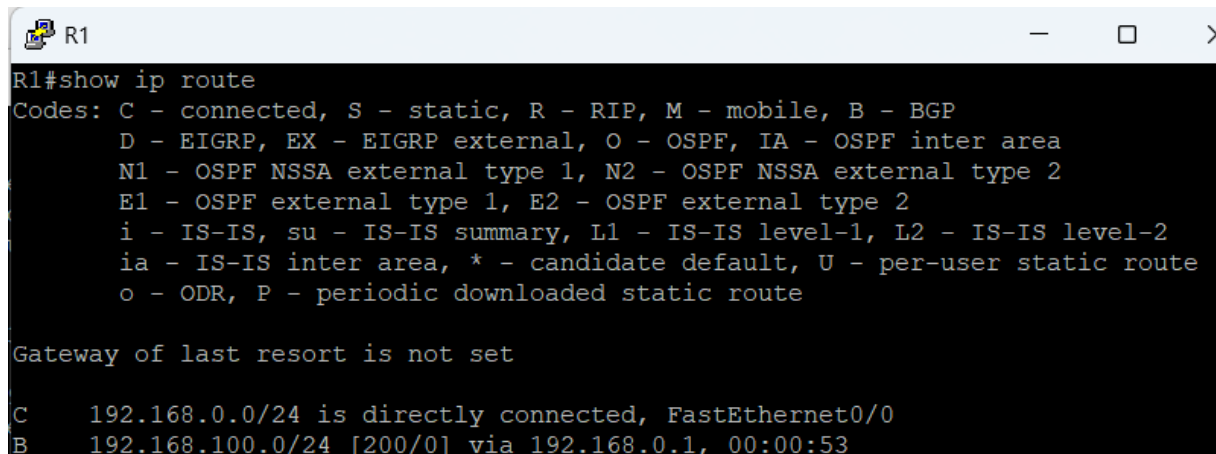
```
R4#conf t
R4(config)#router bgp 65400
R4(config-router)#neighbor 192.168.100.2 next-hop-self
R4(config-router)#exit
```



```
R4
R4#conf t
Enter configuration commands, one per line. End with C
R4(config)#router bgp 65400
R4(config-router)#neighbor 192.168.100.2 next-hop-self
R4(config-router)#exit
```

STEP 7: We execute the “show ip route” command on R1 and R2 to check the routing table.

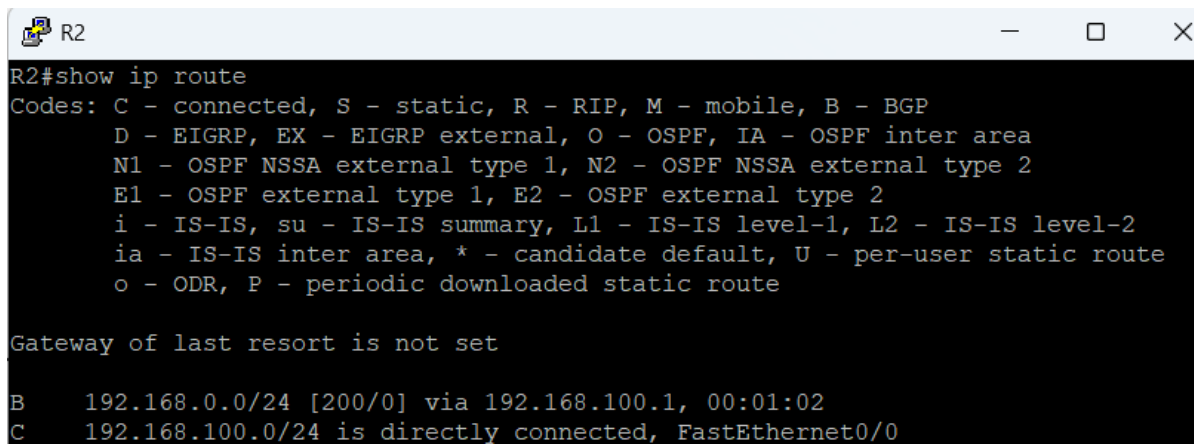
R1#show ip route



```
R1
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.0.0/24 is directly connected, FastEthernet0/0
B    192.168.100.0/24 [200/0] via 192.168.0.1, 00:00:53
```

R2#show ip route


```

R2
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

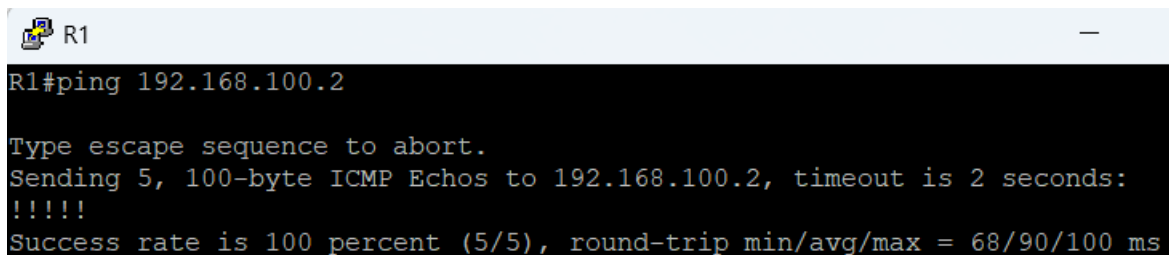
Gateway of last resort is not set

B    192.168.0.0/24 [200/0] via 192.168.100.1, 00:01:02
C    192.168.100.0/24 is directly connected, FastEthernet0/0

```

STEP 8: Now we ping from R1 to R2 and back from R2 to R1.

R1#ping 192.168.100.2



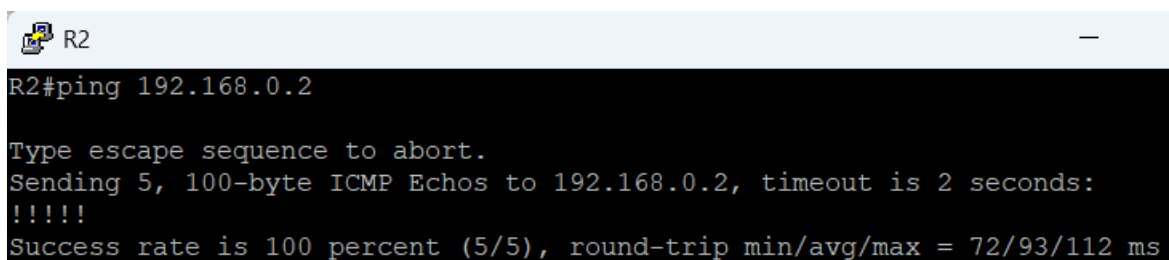
```

R1
R1#ping 192.168.100.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 68/90/100 ms

```

R2#ping 192.168.0.2



```

R2
R2#ping 192.168.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 72/93/112 ms

```

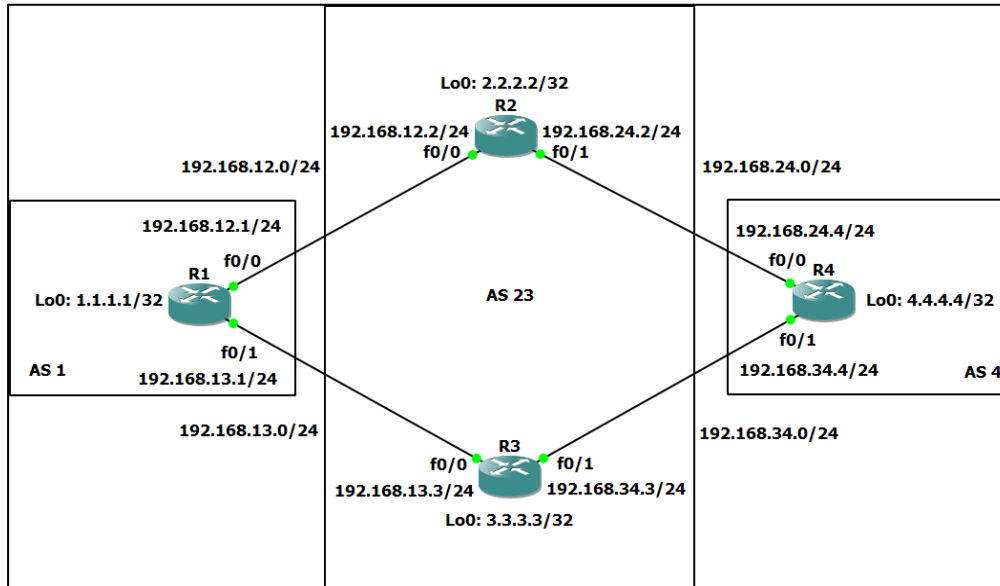
CONCLUSION:

Basic BGP configuration between routers was completed successfully. BGP sessions were established, routes were exchanged, and next-hop issues were resolved using the **next-hop-self** command. After correction, networks became reachable end-to-end, confirming proper BGP operation.

PRACTICAL 4B

AIM: Implementing the concept of BGP Attributes – MED

TOPOLOGY:



STEPS:

STEP 1: Configure interfaces on R1, R2, R3 and R4.

```
R1#conf t
R1(config)#interface Loopback0
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#exit
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit R1(config)#interface
FastEthernet0/1
R1(config-if)#ip address 192.168.13.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

R1

```
R1#conf t
Enter configuration commands, one per line. End with
R1(config)#interface Loopback0
```

```

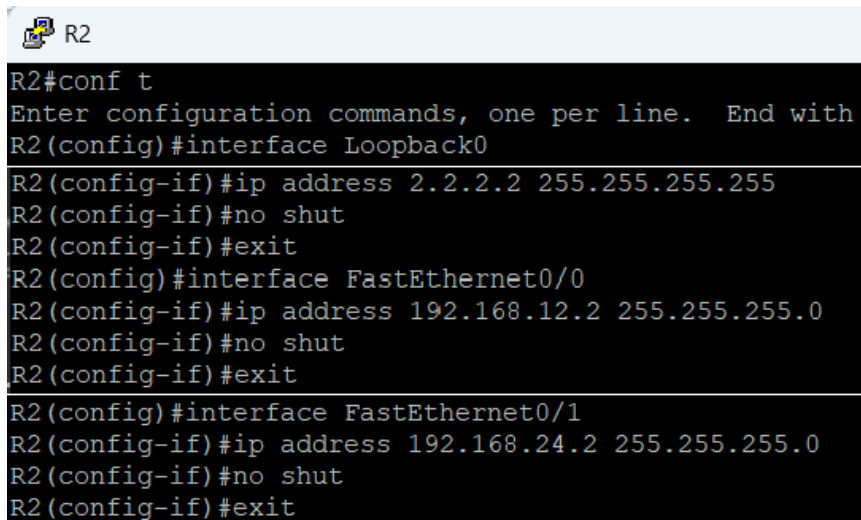
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#exit
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface FastEthernet0/1
R1(config-if)#ip address 192.168.13.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit

```

```

R2#conf t
R2(config)#interface Loopback0
R2(config-if)#ip address 2.2.2.2 255.255.255.255
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit R2(config)#interface
FastEthernet0/1
R2(config-if)#ip address 192.168.24.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit

```



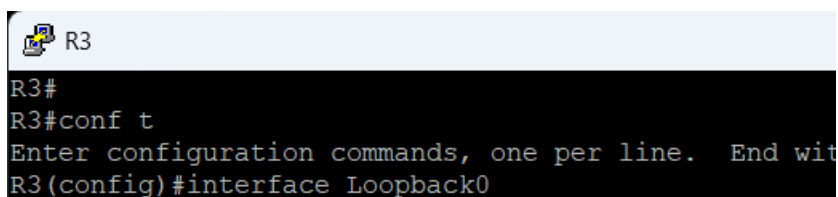
A screenshot of a terminal window titled 'R2' showing the configuration of a Cisco router. The terminal output is as follows:

```

R2#conf t
Enter configuration commands, one per line. End with
R2(config)#interface Loopback0
R2(config-if)#ip address 2.2.2.2 255.255.255.255
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface FastEthernet0/1
R2(config-if)#ip address 192.168.24.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit

```

Router 3:



A screenshot of a terminal window titled 'R3' showing the configuration of a Cisco router. The terminal output is as follows:

```

R3#
R3#conf t
Enter configuration commands, one per line. End with
R3(config)#interface Loopback0

```

```
R3(config-if)#ip address 3.3.3.3 255.255.255.255
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 192.168.13.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

```
R3(config)#interface FastEthernet0/1
R3(config-if)#ip address 192.168.34.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

Router 4:

```
R4
R4#conf t
Enter configuration commands, one per line. End with
R4(config)#interface Loopback0
R4(config-if)#ip address 4.4.4.4 255.255.255.255
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#interface FastEthernet0/0
R4(config-if)#ip address 192.168.24.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#interface FastEthernet0/1
R4(config-if)#ip address 192.168.34.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
```

STEP 2: We execute the "show ip interface brief" command to check if all the interfaces are configured properly.

```
R1
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          192.168.12.1    YES manual up          up
Serial0/0                 unassigned      YES unset   administratively down down
FastEthernet0/1          192.168.13.1    YES manual up          up
```

```
R2
R2#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          192.168.12.2    YES manual up          up
Serial0/0                 unassigned      YES unset   administratively down down
FastEthernet0/1          192.168.24.2    YES manual up          up
```

 R3

```
R3#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Prot
FastEthernet0/0	192.168.13.3	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	192.168.34.3	YES	manual	up	up

 R4

```
R4#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Prot
FastEthernet0/0	192.168.24.4	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	192.168.34.4	YES	manual	up	up

STEP 3: After we configure IP addresses in the interfaces, we configure BGP. The configuration of BGP starts with “router bgp AS” which enters us to the BGP configuration mode. The next step is to configure bgp neighbors with “neighbor (ip- address | peer-group-name) remote-as AS” command. We set neighbors based on ip address.

 R1

```
R1#show running-config | section router bgp
R1#conf t
Enter configuration commands, one per line. End with
R1(config)#router bgp 1
R1(config-router)#no synchronization
R1(config-router)#bgp log-neighbor-changes
R1(config-router)#redistribute connected
R1(config-router)#neighbor 192.168.12.2 remote-as 23
R1(config-router)#neighbor 192.168.13.3 remote-as 23
R1(config-router)#no auto-summary
R1(config-router)#end
```

 R2

```
R2#show running-config | section router bgp
R2#conf t
Enter configuration commands, one per line. End with
R2(config)#router bgp 23
R2(config-router)#no synchronization
R2(config-router)#bgp log-neighbor-changes
R2(config-router)#redistribute connected
R2(config-router)#neighbor 192.168.12.1 remote-as 1
R2(config-router)#neighbor 192.168.24.4 remote-as 4
```

```
R3
R3#show running-config | section router bgp
R3#conf t
Enter configuration commands, one per line.  End with
R3(config)#router bgp 23
R3(config-router)#no synchronization
R3(config-router)#bgp log-neighbor-changes
R3(config-router)#redistribute connected
R3(config-router)#neighbor 192.168.13.1 remote-as 1
R3(config-router)#neighbor 192.168.34.4 remote-as 4
R3(config-router)#no auto-summary
R3(config-router)#end
```

```
R4
R4#show running-config | section router bgp
R4#conf t
Enter configuration commands, one per line.  End with
R4(config)#router bgp 4
R4(config-router)#no synchronization
R4(config-router)#bgp log-neighbor-changes
R4(config-router)#redistribute connected
R4(config-router)#neighbor 192.168.24.2 remote-as 23
R4(config-router)#neighbor 192.168.34.3 remote-as 23
R4(config-router)#no auto-summary
*Mar  1 00:16:08.107: %BGP-5-ADJCHANGE: neighbor 192.
R4(config-router)#no auto-summary
R4(config-router)#end
```

STEP 4: From the below R1 BGP table, we can see that the R1 prefers the path through R3 (Next Hop IP as 192.168.13.3) to reach the Network 4.4.4.4/32.

Symbol ">" in-front of the network indicates the best path (router prefers to use this path to send the traffic Now, our task to change the path through R2 to reach the Network 4.4.4.4/32.


```

R1
R1#show ip bgp
BGP table version is 12, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*> 1.1.1.1/32      0.0.0.0                0           32768 ?
*> 2.2.2.2/32      192.168.12.2            0             0 23 ?
*> 3.3.3.3/32      192.168.13.3            0             0 23 ?
*  4.4.4.4/32      192.168.13.3            0             0 23 4 ?
*>                 192.168.12.2            0             0 23 4 ?
* 192.168.12.0     192.168.12.2            0             0 23 ?
*>                 0.0.0.0                0           32768 ?
* 192.168.13.0     192.168.13.3            0             0 23 ?
*>                 0.0.0.0                0           32768 ?
* 192.168.24.0     192.168.13.3            0             0 23 4 ?
*>                 192.168.12.2            0             0 23 ?
* 192.168.34.0     192.168.12.2            0             0 23 4 ?
*>                 192.168.13.3            0             0 23 ?

```

STEP 5: Now we create a route-map on R3.

```

R3
R3#conf t
Enter configuration commands, one per line
R3(config)#route-map MED-TEST permit
R3(config-route-map)#match ip address 1
R3(config-route-map)#set metric 500
R3(config-route-map)#exit
R3(config)#route-map MED-TEST permit 20
R3(config-route-map)#end

```

STEP 6: We can also view the created route-map using the "show route-map" command.

```

R3
R3#show route-map
route-map MED-TEST, permit, sequence 10
  Match clauses:
    ip address (access-lists): 1
  Set clauses:
    metric 500
  Policy routing matches: 0 packets, 0 bytes
route-map MED-TEST, permit, sequence 20
  Match clauses:
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes

```

STEP 7: Next we create an access-list as.

```

R3
R3#conf t
Enter configuration commands, one per line.
R3(config)#access-list 1 permit host 4.4.4.4
R3(config)#end

```

STEP 8: We can view the configured access-list using the "show access-list" command.

```

R3
R3#show access-lists
Standard IP access list 1
  10 permit 4.4.4.4

```

STEP 9: Now we apply the created route-map to the bgp configuration. We map the route-map MED-TEST as "OUT" to the neighbor R1 (192.168.13.1). so that it will advertise the configured Metric value (value: 500) its neighbor.

```

R3
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router bgp 23
R3(config-router)#neighbor 192.168.13.1 route-map MED-TEST out
R3(config-router)#end

```

STEP 10: We clear the BGP session so that the newly applied configuration takes effect. In-order to clear the bgp session, we use the command "clear ip bgp * soft in" and then we execute the "show ip bgp" command again.

```

R1
R1#clear ip bgp * soft in
R1#show ip bgp
BGP table version is 12, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*>  1.1.1.1/32     0.0.0.0               0         32768 ?
*>  2.2.2.2/32     192.168.12.2          0         0 23 ?
*>  3.3.3.3/32     192.168.13.3          0         0 23 ?
*   4.4.4.4/32     192.168.13.3          500        0 23 4 ?
*>  192.168.12.0   192.168.12.2          0         0 23 4 ?
*>  192.168.12.0   192.168.12.2          0         0 23 ?
*>  192.168.13.0   192.168.13.3          0         0 23 ?
*>  192.168.13.0   192.168.13.3          0         0 23 ?
*>  192.168.24.0   192.168.13.3          0         0 23 4 ?
*>  192.168.24.0   192.168.12.2          0         0 23 ?
*>  192.168.34.0   192.168.12.2          0         0 23 4 ?
*>  192.168.34.0   192.168.13.3          0         0 23 ?

```

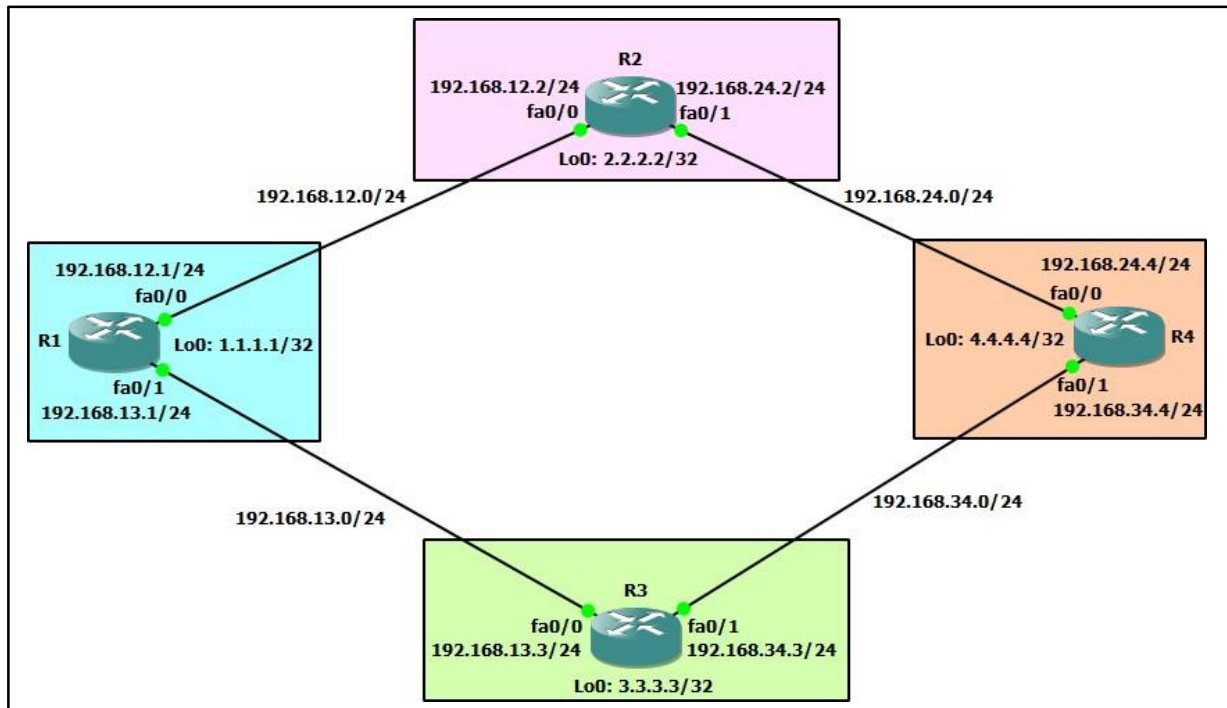
CONCLUSION :

The MED attribute was applied successfully using route-maps to influence inbound path selection. After applying the MED value and clearing BGP sessions, the preferred path changed as intended. This confirmed the effect of the MED attribute on traffic engineering in BGP.

PRACTICAL 4C

AIM: Implement the concept of BGP AS Path Attribute

TOPOLOGY:



STEPS:

STEP 1: Configure interfaces on R1, R2, R3 and R4.

Router 1

```

R1
R1#conf t
Enter configuration commands, one per line. End with
R1(config)#interface Loopback0
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#exit
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface FastEthernet0/1
R1(config-if)#ip address 192.168.13.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
  
```

Router 2

```
R2
R2#conf t
Enter configuration commands, one per line.  End with Ctrl-Z to exit.
R2(config)#interface Loopback0
R2(config-if)#ip address 2.2.2.2 255.255.255.255
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface FastEthernet0/1
R2(config-if)#ip address 192.168.24.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#end
```

Router 3

```
R3
R3#conf t
Enter configuration commands, one per line.  End with Ctrl-Z to exit.
R3(config)#interface Loopback0
R3(config-if)#ip address 3.3.3.3 255.255.255.255
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 192.168.13.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#interface FastEthernet0/1
R3(config-if)#ip address 192.168.34.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#end
```

Router 4

```
R4
R4#conf t
Enter configuration commands, one per line.  End with Ctrl-Z to exit.
R4(config)#interface Loopback0
R4(config-if)#ip address 4.4.4.4 255.255.255.255
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#interface FastEthernet0/0
R4(config-if)#ip address 192.168.24.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#interface FastEthernet0/1
R4(config-if)#ip address 192.168.34.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#end
```

Step 2: We execute the "show ip interface brief" command to check if all the interfaces

are configured properly.

R1					
R1#show ip interface brief					
Interface	IP-Address	OK?	Method	Status	Prot
FastEthernet0/0	192.168.12.1	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	192.168.13.1	YES	manual	up	up

R2					
R2#show ip interface brief					
Interface	IP-Address	OK?	Method	Status	Prot
FastEthernet0/0	192.168.12.2	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	192.168.24.2	YES	manual	up	up

R3					
R3#show ip interface brief					
Interface	IP-Address	OK?	Method	Status	Prot
FastEthernet0/0	192.168.13.3	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	192.168.34.3	YES	manual	up	up

R4					
R4#show ip interface brief					
Interface	IP-Address	OK?	Method	Status	Prot
FastEthernet0/0	192.168.24.4	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	192.168.34.4	YES	manual	up	up

Step 3: After we configure IP addresses in the interfaces, we configure BGP. The configuration of BGP starts with “router bgp AS” which enters us to the BGP configuration mode. The next step is to configure bgp neighbors with “neighbor (ip-address | peer-group-name) remote-as AS” command. We set neighbors based on ip address.

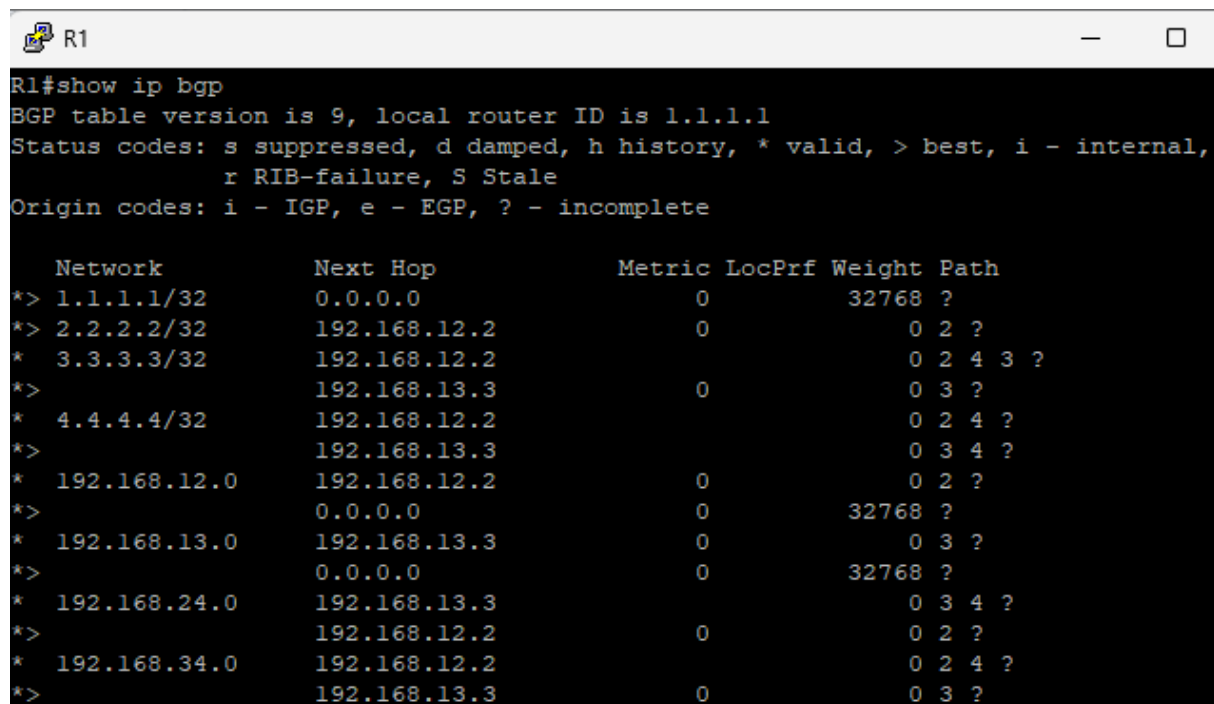
```
R1
R1#conf t
Enter configuration commands, one per line. End with
R1(config)#end
R1#
*Mar 1 00:15:06.363: %SYS-5-CONFIG_I: Configured from
R1#show running-config | section router bgp
R1#conf t
Enter configuration commands, one per line. End with
R1(config)#router bgp 1
R1(config-router)# no synchronization
R1(config-router)# bgp log-neighbor-changes
R1(config-router)# redistribute connected
R1(config-router)# neighbor 192.168.12.2 remote-as 2
R1(config-router)# neighbor 192.168.13.3 remote-as 3
R1(config-router)# no auto-summary
```

```
R2
R2#show running-config | section router bgp
R2#conf t
Enter configuration commands, one per line. End with
R2(config)#router bgp 2
R2(config-router)# no synchronization
R2(config-router)# bgp log-neighbor-changes
R2(config-router)# redistribute connected
R2(config-router)# neighbor 192.168.12.1 remote-as 1
R2(config-router)# neighbor 192.168.24.4 remote-as 4
R2(config-router)# no auto-summary
```

```
R3
R3#show running-config | section router bgp
R3#conf t
Enter configuration commands, one per line. End with
R3(config)#router bgp 3
R3(config-router)# no synchronization
R3(config-router)# bgp log-neighbor-changes
R3(config-router)# redistribute connected
R3(config-router)# neighbor 192.168.13.1 remote-as 1
R3(config-router)# neighbor 192.168.34.4 remote-as 4
R3(config-router)# no auto-summary
```

```
R4
R4#show running-config | section router bgp
R4#conf t
Enter configuration commands, one per line. End with
R4(config)#router bgp 4
R4(config-router)# no synchronization
R4(config-router)# bgp log-neighbor-changes
R4(config-router)# redistribute connected
R4(config-router)# neighbor 192.168.24.2 remote-as 2
R4(config-router)# neighbor 192.168.34.3 remote-as 3
R4(config-router)# no auto-summary
```

STEP 4: From the table below, network 4.4.4.4/32 has two paths R2 and R3. Next-hop IP 192.168.12.2 refers to R2 and next-hop IP 192.168.13.3 refers to R3. R1 prefers R2 (Next hop - 192.168.12.2) to reach the destination.



```

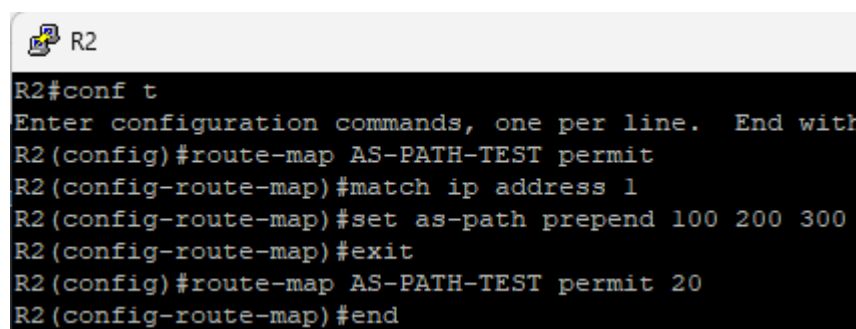
R1#show ip bgp
BGP table version is 9, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop        Metric LocPrf Weight Path
*> 1.1.1.1/32      0.0.0.0          0           32768 ?
*> 2.2.2.2/32      192.168.12.2      0           0 2 ?
* 3.3.3.3/32      192.168.12.2      0           0 2 4 3 ?
*>                192.168.13.3      0           0 3 ?
* 4.4.4.4/32      192.168.12.2      0           0 2 4 ?
*>                192.168.13.3      0           0 3 4 ?
* 192.168.12.0    192.168.12.2      0           0 2 ?
*>                0.0.0.0           0          32768 ?
* 192.168.13.0    192.168.13.3      0           0 3 ?
*>                0.0.0.0           0          32768 ?
* 192.168.24.0    192.168.13.3      0           0 3 4 ?
*>                192.168.12.2      0           0 2 ?
* 192.168.34.0    192.168.12.2      0           0 2 4 ?
*>                192.168.13.3      0           0 3 ?

```

Best path is indicated using ">" symbol. Now we start configuring AS Path prepend to make R3 as the best path from R1 for the Network 4.4.4.4/32.

STEP 5: We have created a route-map named as AS-PATH-TEST. Next we use the command "match ip address 1" to match the access-list 1, and then if it matches we set the as-path prepend as 100 200 300. We then create a second permit statement for the same route-map AS-PATH-TEST.

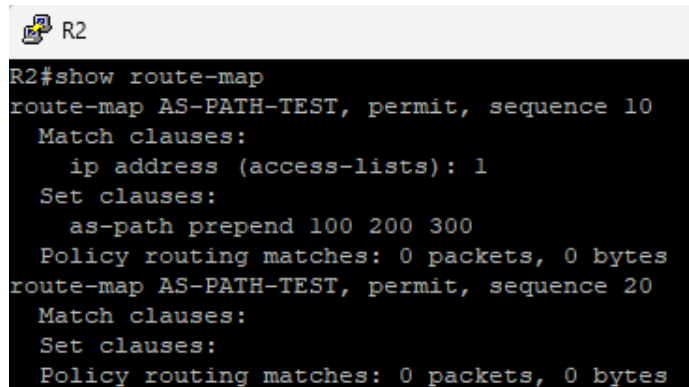


```

R2#conf t
Enter configuration commands, one per line. End with
R2(config)#route-map AS-PATH-TEST permit
R2(config-route-map)#match ip address 1
R2(config-route-map)#set as-path prepend 100 200 300
R2(config-route-map)#exit
R2(config)#route-map AS-PATH-TEST permit 20
R2(config-route-map)#end

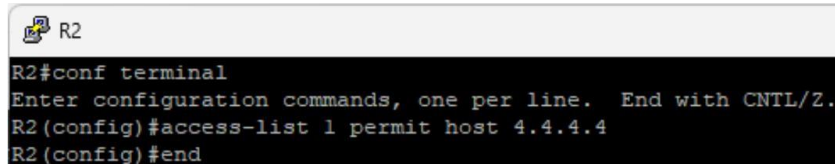
```


STEP 6: We can also view the created route-map using the "show route-map" command.



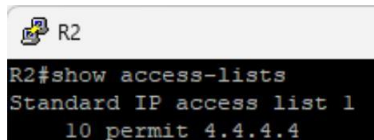
```
R2
R2#show route-map
route-map AS-PATH-TEST, permit, sequence 10
  Match clauses:
    ip address (access-lists): 1
  Set clauses:
    as-path prepend 100 200 300
  Policy routing matches: 0 packets, 0 bytes
route-map AS-PATH-TEST, permit, sequence 20
  Match clauses:
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
```

STEP 7: Next we create an access-list as.



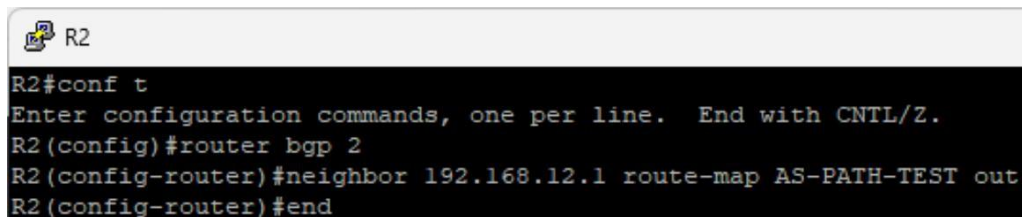
```
R2
R2#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#access-list 1 permit host 4.4.4.4
R2(config)#end
```

STEP 8: We can view the configured access-list using the "show access-list" command.



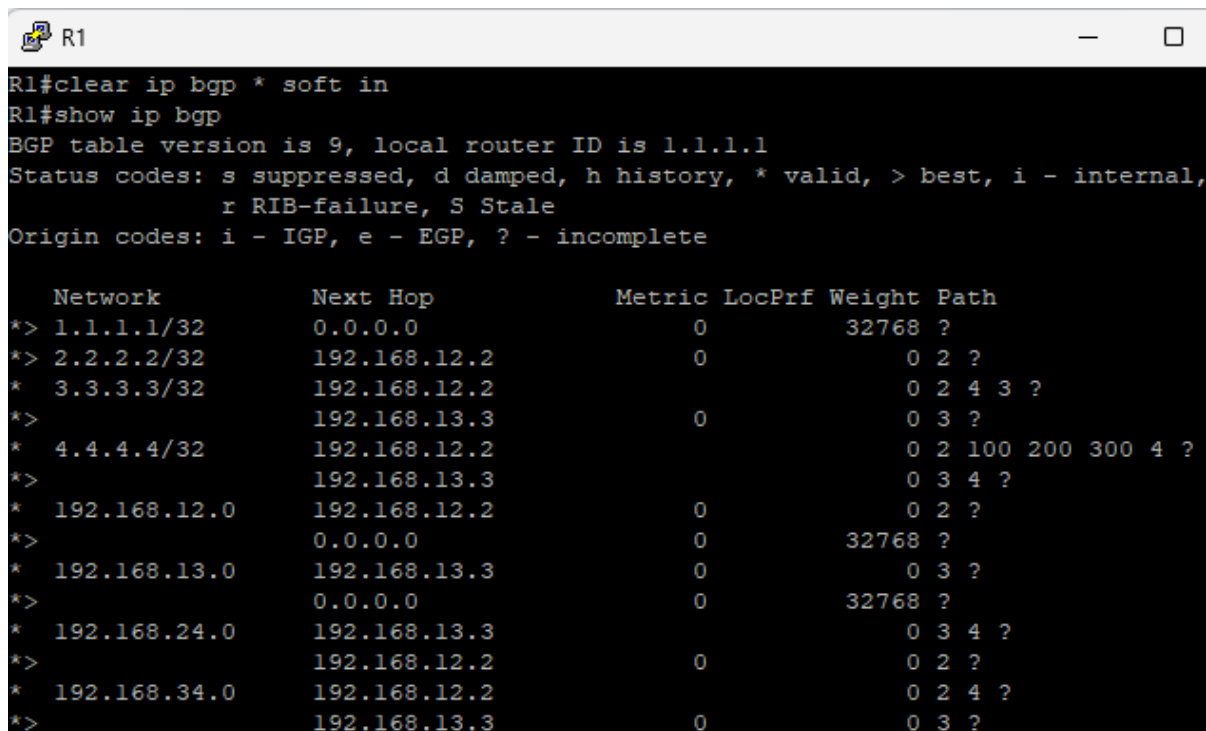
```
R2
R2#show access-lists
Standard IP access list 1
  10 permit 4.4.4.4
```

STEP 9: Now we apply the created route-map to the bgp configuration. We map the route-map AS-PATH-TEST as "OUT" to the neighbor R1 (192.168.12.1) so that it will advertise the additional AS number to its neighbor.



```
R2
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router bgp 2
R2(config-router)#neighbor 192.168.12.1 route-map AS-PATH-TEST out
R2(config-router)#end
```

STEP 10: We clear the BGP session so that the newly applied configuration takes effect. In-order to clear the bgp session, we use the command "clear ip bgp * soft in" and then we execute the "show ip bgp" command again.



```

R1#clear ip bgp * soft in
R1#show ip bgp
BGP table version is 9, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop        Metric LocPrf Weight Path
*> 1.1.1.1/32      0.0.0.0              0           32768 ?
*> 2.2.2.2/32      192.168.12.2         0           0 2 ?
* 3.3.3.3/32      192.168.12.2         0           0 2 4 3 ?
*>                192.168.13.3         0           0 3 ?
* 4.4.4.4/32      192.168.12.2         0           0 2 100 200 300 4 ?
*>                192.168.13.3         0           0 3 4 ?
* 192.168.12.0    192.168.12.2         0           0 2 ?
*>                0.0.0.0              0           32768 ?
* 192.168.13.0    192.168.13.3         0           0 3 ?
*>                0.0.0.0              0           32768 ?
* 192.168.24.0    192.168.13.3         0           0 3 4 ?
*>                192.168.12.2         0           0 2 ?
* 192.168.34.0    192.168.12.2         0           0 2 4 ?
*>                192.168.13.3         0           0 3 ?

```

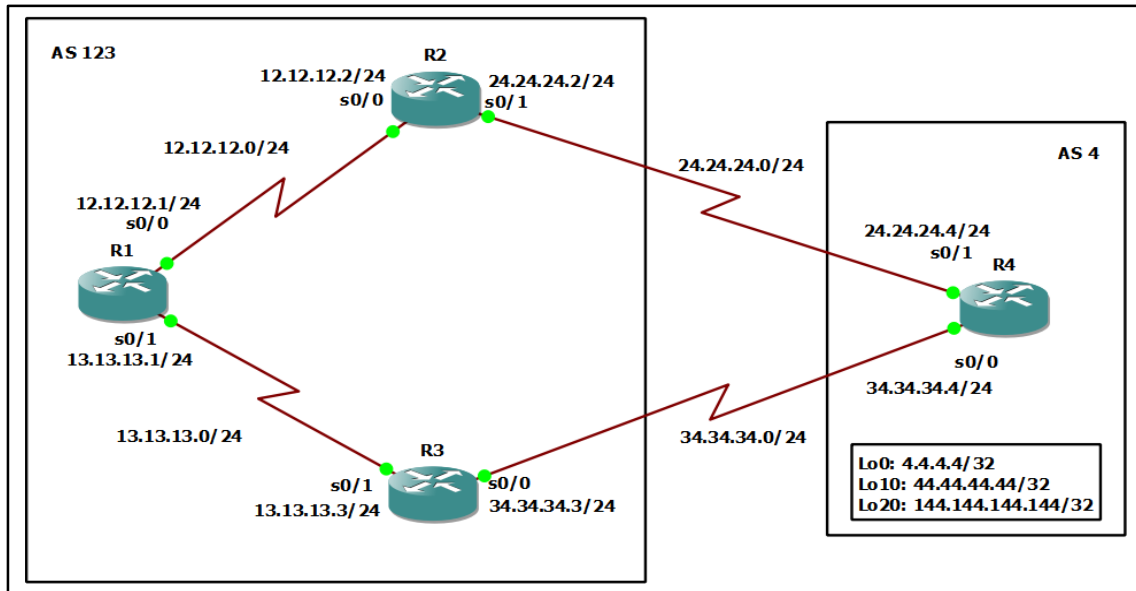
CONCLUSION:

AS Path Prepending was implemented successfully to manipulate outbound traffic paths. The modified AS path made the alternate link more preferred from the neighbor's perspective. Route-maps and access lists were applied correctly, and BGP tables confirmed the updated path selection.

PRACTICAL 4D

AIM: Implementing the concept of BGP Path Attribute – Local Preference

TOPOLOGY:



STEPS:

STEP 1: Configure interfaces on R1, R2, R3 and R4.

ROUTER 1

```

R1
R1#conf t
Enter configuration commands, one per line. End with Ctrl-Z.
R1(config)#interface Serial0/0
R1(config-if)#ip address 12.12.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#interface Serial0/1
R1(config-if)#ip address 13.13.13.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#end

```

 R2

```
R2#conf t
Enter configuration commands, one per line. End with CNF
R2(config)#interface Serial0/0
R2(config-if)#ip address 12.12.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit

R2(config)#interface Serial0/1
R2(config-if)#ip address 24.24.24.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#end
```

 R3

```
R3#conf t
Enter configuration commands, one per line. End with CNF
R3(config)#interface Serial0/0
R3(config-if)#ip address 34.34.34.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit

R3(config)#interface Serial0/1
R3(config-if)#ip address 13.13.13.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#end
```

 R4

```
R4#conf t
Enter configuration commands, one per line. End with CNF
R4(config)#interface Serial0/0
R4(config-if)#ip address 34.34.34.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit

R4(config)#interface Serial0/1
R4(config-if)#ip address 24.24.24.4 255.255.255.0
R4(config-if)#no shut
R4(config-if)#exit

R4(config)#interface Loopback0
R4(config-if)#ip address 4.4.4.4 255.255.255.255
R4(config-if)#no shut
R4(config-if)#exit

R4(config)#interface Loopback10
R4(config-if)#ip address 44.44.44.44 255.255.255.255
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#interface Loopback20
R4(config-if)#ip address 144.144.144.144 255.255.255.255
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#end
```

STEP 2: We execute the "show ip interface brief" command to check if all the interfaces are configured properly.

 R1

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
Serial0/0	12.12.12.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/1	13.13.13.1	YES	manual	up	up

 R2

```
R2#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
Serial0/0	12.12.12.2	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/1	24.24.24.2	YES	manual	up	up

 R3

```
R3#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
Serial0/0	34.34.34.3	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/1	13.13.13.3	YES	manual	up	up

 R4

```
R4#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
Serial0/0	34.34.34.4	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/1	24.24.24.4	YES	manual	up	up
Loopback0	4.4.4.4	YES	manual	up	up
Loopback10	44.44.44.44	YES	manual	up	up
Loopback20	144.144.144.144	YES	manual	up	up

STEP 3: After we configure IP addresses in the interfaces, we configure BGP. The configuration of BGP starts with "router bgp AS" which enters us to the BGP configuration mode. The next step is to configure bgp neighbors with "neighbor (ip-address | peer-group-name) remote-as AS" command. We set neighbors based on ip address.

 R1

```
R1#conf t
Enter configuration commands, one per line. End with
R1(config)#router bgp 123
R1(config-router)#no synchronization
R1(config-router)#bgp log-neighbor-changes
R1(config-router)#neighbor 12.12.12.2 remote-as 123
R1(config-router)#neighbor 13.13.13.3 remote-as 123
R1(config-router)#no auto-summary
R1(config-router)#end
```

 R2

```
R2#conf t
Enter configuration commands, one per line. End with
R2(config)#router bgp 123
R2(config-router)#no synchronization
R2(config-router)#bgp log-neighbor-changes
R2(config-router)#neighbor 12.12.12.1 remote-as 123
R2(config-router)#neighbor 12.12.12.1 next-hop-self
R2(config-router)#neighbor 24.24.24.4 remote-as 4
R2(config-router)#no auto-summary
```

 R3

```
R3#conf t
Enter configuration commands, one per line. End with
R3(config)#router bgp 123
R3(config-router)#no synchronization
R3(config-router)#bgp log-neighbor-changes
R3(config-router)#neighbor 13.13.13.1 remote-as 123
R3(config-router)#neighbor 13.13.13.1 next-hop-self
R3(config-router)#neighbor 34.34.34.4 remote-as 4
R3(config-router)#no auto-summary
R3(config-router)#end
```

 R4

```
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router bgp 4
R4(config-router)#no synchronization
R4(config-router)#bgp log-neighbor-changes
R4(config-router)#network 4.4.4.4 mask 255.255.255.255
R4(config-router)#network 44.44.44.44 mask 255.255.255.255
R4(config-router)#network 144.144.144.144 mask 255.255.255.255
R4(config-router)#neighbor 24.24.24.2 remote-as 123
R4(config-router)#neighbor 34.34.34.3 remote-as 123
R4(config-router)#no auto-summary
R4(config-router)#end
```

Step 4: We use the “sh ip bgp” command to check which is the best path taken by R1 to reach the networks advertised by R4.

 R1

```
R1#sh ip bgp
BGP table version is 7, local router ID is 13.13.13.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop           Metric LocPrf Weight Path
*>i4.4.4.4/32       12.12.12.2             0      100      0 4 i
* i                 13.13.13.3             0      100      0 4 i
*>i44.44.44.44/32   12.12.12.2             0      100      0 4 i
* i                 13.13.13.3             0      100      0 4 i
*>i144.144.144.144/32
12.12.12.2          0      100      0 4 i
* i                 13.13.13.3             0      100      0 4 i
```

STEP 5: It is clear to us that it prefers to take R2 to reach the networks in R4. We now configure R3 so that all routes received by R3 will have a Local Preference of 300.

```

R3
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router bgp 123
R3(config-router)#bgp default local-preference 300
R3(config-router)#end

```

STEP 6: Now without clearing the BGP process, R3 now became the more preferred path to reach R4. In Local Preference, the higher the value, the more it is preferred.

```

R1
R1#sh ip bgp
BGP table version is 10, local router ID is 13.13.13.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop        Metric LocPrf Weight Path
*  i4.4.4.4/32     12.12.12.2         0      100      0 4 i
*>i               13.13.13.3         0      300      0 4 i
* i44.44.44.44/32  12.12.12.2         0      100      0 4 i
*>i               13.13.13.3         0      300      0 4 i
* i144.144.144.144/32
   12.12.12.2         0      100      0 4 i
*>i               13.13.13.3         0      300      0 4 i

```

STEP 7: We now configure a route-map in R2 so that the network 144.144.144.144/32 will have a Local Preference of 500. This will make R2 the best path to reach the mentioned network.

```

R2
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#access-list 1 permit host 144.144.144.144
R2(config)#route-map LOCALPREF500 permit 10
R2(config-route-map)#match ip address 1
R2(config-route-map)#set local-preference 500
R2(config-route-map)#exit
R2(config)#router bgp 123
R2(config-router)#neighbor 24.24.24.4 route-map LOCALPREF500 in
R2(config-router)#exit
R2(config)#exit

```

STEP 8: We use the “sh ip bgp” command to check which is the best path NOW taken by R1 to reach the networks advertised by R4. We see that the Local Preference has been set.

```

R1
R1#sh ip bgp
BGP table version is 11, local router ID is 13.13.13.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*>i4.4.4.4/32      13.13.13.3             0     300      0 4 i
*>i44.44.44.44/32  13.13.13.3             0     300      0 4 i
*>i144.144.144.144/32
* i               13.13.13.3             0     300      0 4 i

```

STEP 9: We configure the 20th sequence of the route-map on R2.

```

R2
R2#conf t
Enter configuration commands, one per line.
R2(config)#route-map LOCALPREF500 permit 20
R2(config-route-map)#end

```

STEP 10: Let us see if there are any changes

```

R1
R1#sh ip bgp
BGP table version is 11, local router ID is 13.13.13.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*>i4.4.4.4/32      13.13.13.3             0     300      0 4 i
*>i44.44.44.44/32  13.13.13.3             0     300      0 4 i
*>i144.144.144.144/32
* i               13.13.13.3             0     300      0 4 i

```

CONCLUSION:

Local Preference was configured to control outbound traffic within the AS. Increasing the Local Preference on selected routers successfully influenced path selection towards preferred links. The BGP table updates validated that routers correctly chose the highest Local Preference path, demonstrating effective inbound/outbound path engineering.